AQ eReporting Guide

Schema version: 5.02 – June 2026
Reporting guide version: R3_v502_AQ_Reporting_guide_v2 – June 2026

Welcome to the AQ eReporting Guide.

This guide provides structured documentation for the Reportnet 3 air quality reporting tables, their attributes, identifiers, relationships and examples.



Introduction

Scope, purpose and general principles of the guide.



Diagram

Overview of the reporting structure and table relationships.



Tables

Detailed documentation of reporting tables and attributes.



Identifiers

Reference information on identifiers, keys and links.



Table relationships

Physical and logical relationships between reporting tables.

Introduction

Re-Submission and submission control through legacy/reference data

The data will be provided within 4 data flows:

comprehensive AQ data flow working in ‘continuous’ (re)submission mode (ex all data flows),

compliance validation with deadline of 30 of September (ex B to G data flows),

AQ plans validation with deadline of 31 of December (ex H to K data flows),

confirmation of provisional assessment regimes with deadline of 31 of December (ex pB and pC data flows).

Comprehensive AQ data flow working in ‘continuous’ (re)submission mode

This is the only data flow where the actual reporting of data takes place.

All data should be allowed for submissions at any time, providing that they passed QC. The same applies for resubmissions.

The countries have to submit well in advance all data necessary for validation and confirmation at deadlines. This is to allow for data processing at the EEA, so that the tools provided by the Agency for the purpose of the validation and/or confirmation are ready on time.

The data submission should allow ‘management by change’, i.e. only reporting data which is new.

Validation and confirmation at deadlines

The idea is that this will not be ‘real’ reporting data flow but will only require the country to validate/confirm what EEA concludes from data submitted within the ‘Comprehensive AQ data flow’.

All tables will be pre-filled, based on information collected so far and countries will be asked to validate it by releasing.

The pre-filled tables will not be editable.

The pre-filling could be happening ‘real time’ and be visible for countries as they deliver data in the ‘Comprehensive AQ data flow’ - with particular emphasis on `ComplianceAssessmentMethod` and `CompliancePlanLink` tables. However, the validation/confirmation data flows will open for release for period of 30 - 45 days before the deadline. This will also allow countries to validate/confirm their resubmissions.

The validation/confirmation at deadlines will be allowed (Technical Acceptance) only if `ComplianceAssessmentMethod` and `CompliancePlanLink` tables are complete or if their incompleteness does not originate from technical issues with data.

There will be other tools (outside R3) provided for countries for assessing completeness and quality of their data (re)submitted in the ‘Comprehensive AQ data flow’.

These external tools as well as the pre-filling will be fed from EEA’s data base collecting data (re)submitted in the ‘Comprehensive AQ data flow’ and integrating it with the historical information.

EEA will have a parallel database with a snapshot of the legacy/reference data set taken at each deadline. It will serve as the main source of data for products as well as reference to compare with data processed after resubmissions. The resubmitted data will be allowed for validation/confirmation only if they do not affect previous compliance status.

Reference data set

The reference data set will be wider than legacy data from AQ e-Reporting, it will include:

legacy data from AQ e-Reporting,

vocabularies,

gridded NUTS regions (100m) - to be generated by the EEA,

gridded AQ zones (100/1000 m) - to be generated by the EEA (and a process to automatically grid new zone geometries should be established),

gridded E1b results from the current e-Reporting system - to be generated by the EEA,

gridded sampling point locations - to be generated by the EEA,

gridded SRs - European coverage (100m) - to be generated by the EEA.

Legacy data cleansing for the reference data flow

The legacy data collected within the current system will have to be reviewed and cleaned before it can be used for QC as reference.

The best way to achieve clean and consistent data set will be following these steps:

complete work on QC rules, iteratively improving the set of QCs (EEA)

generate full set of legacy data per country and run through the QCs (EEA)

ask country reporters to fix critical issues and upload the corrected legacy data to R3 to run through the QC rules

once this is done successfully, clean and consistent data set should be established for each country

Maintenance of spatial coherence using common grid

The spatial coherence of the data will be maintained through common grid approach.

The common grid approach is understood as use of the EEA INSPIRE grid in Europe LAEA (SRID 3035) projection and recommended resolution steps (10, 100, 1000, 10000 m) for:

locations of sampling points,

zone geometries (reported by countries as polygons and gridded by the EEA),

exceedance extents,

modelling results (to be reported by countries on common grid),

SRs (although harmonised European SRs may be provided by the EEA, it can be overwritten by countries - reporting their own results on common grid).

Reporting in other resolutions will be allowed, fulfilling these conditions:

For resolutions < 100 m: integer multiple of 10, e.g.: 20, 30, 40, 50 m, etc.

For resolutions > 100 m: integer multiple of 100, e.g.: 200, 300, 400, … , 2500 m, 5000 m, etc.

The benefits of using the common grid approach

harmonised spatial reference for several elements: sampling points, representativeness areas, modelling results, exceedance extent (also AQ zones can be represented as grid),

simplified reporting of modelling results (flat table, geometry not needed),

efficient processing of reported data, e.g.: locations of sampling points (in zones, countries, NUTS regions), compliance statistics from modelling results, MQI calculations (sampling points vs modelling results), etc.

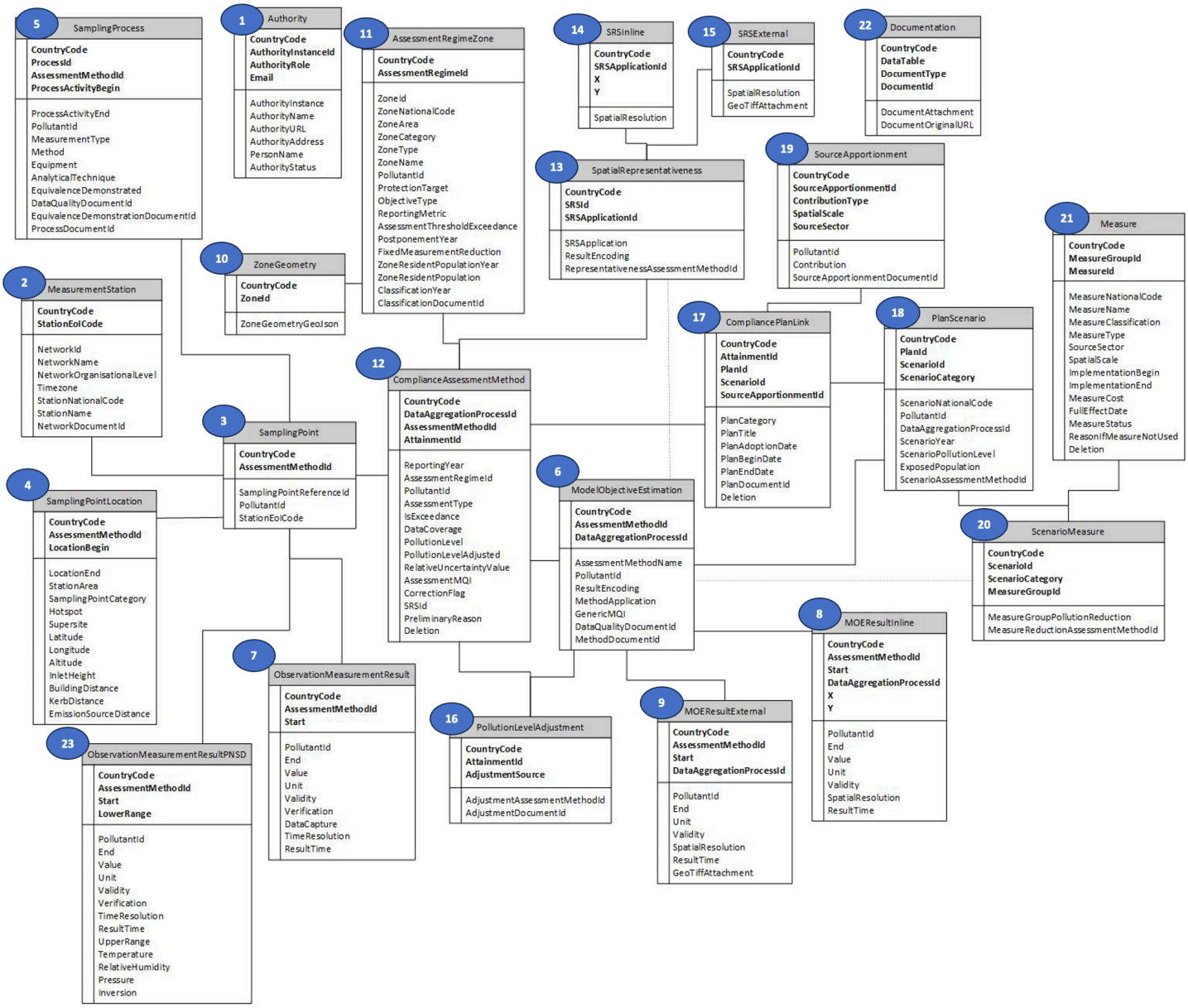
extending operational capacity of AQ e-Reporting, e.g.: processing of data reported for adjustments (natural sources, WSS), possibly - also from scenario modelling,

direct link to EEA’s assessments: common grid with EEA’s data on environmental burden of disease from air pollution, population data, AQ maps,

readiness for products: much less processing efforts needed to deliver products (online viewers, maps, etc).

Data model diagram

The following diagram provides an overview of the Reportnet air quality reporting data model (v502).



Tables shown in the diagram

No	Table
1	Authority
2	MeasurementStation
3	SamplingPoint
4	SamplingPointLocation
5	SamplingProcess

No	Table
6	ModelObjectiveEstimation
7	ObservationMeasurementResult
8	MOEResultInline
9	MOEResultExternal
10	ZoneGeometry
11	AssessmentRegimeZone
12	ComplianceAssessmentMethod
13	SpatialRepresentativeness
14	SRSInline
15	SRSExternal
16	PollutionLevelAdjustment
17	CompliancePlanLink
18	PlanScenario
19	SourceApportionment
20	ScenarioMeasure
21	Measure
22	Documentation
23	ObservationMeasurementResultPNSD

Tables

Select a reporting table below to view its structure, attributes and examples.

Authority



MeasurementStation



SamplingPoint



SamplingPointLocation



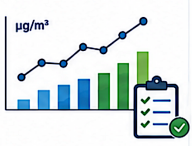
SamplingProcess



ModelObjectiveEstimation



ObservationMeasurementResult



MOEResultInline



MOEResultExternal



AssessmentRegimeZone



ZoneGeometry



ComplianceAssessmentMethod



SpatialRepresentativeness



SRSInline



SRSExternal



PollutionLevelAdjustment



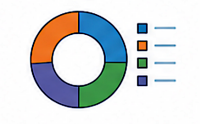
CompliancePlanLink



PlanScenario



SourceApportionment



ScenarioMeasure



Measure



Documentation



ObservationMeasurementResultPNSD



Authority



Description

EEA extensions

Data management rules

Attributes

Attribute details

AUT_01 – CountryCode

AUT_02 – AuthorityInstanceId

AUT_03 – AuthorityRole

AUT_04 – Email

AUT_05 – AuthorityInstance

AUT_06 – AuthorityName

AUT_07 – AuthorityURL

AUT_08 – AuthorityAddress

AUT_09 – PersonName

AUT_10 – AuthorityStatus

Example

Description

The purpose of the Authority table is to communicate the contact details of the responsible authorities. It describes the authorities and their area of responsibility (AuthorityRole).

Authority table has to be reported whenever new information or an update of existing information is available.

Attributes which create unique identifier of each record: CountryCode, AuthorityInstanceId, AuthorityRole and Email.

Updates of the Authority table will be managed using reference/legacy data, where each addition and modification will be stamped with ReportingTime:

addition of new record is understood as addition of new combination of values for the attributes creating unique identifier,

modification of existing record is understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC rules to be established, e.g. cross-checking AuthorityInstanceId against Zone table or MeasurementStation table.

Valid modifications of existing record in this table are almost all allowed, only correction of AuthorityInstance may be restricted as it depends on AuthorityInstanceId. Validity of the modifications of existing record will be tested using QC rules to be established. Allowed modifications are not supposed to affect outcomes of compliance analysis and/or assessments.

Data products for public will present the records with active AuthorityStatus and the latest ReportingTime for distinct combination of CountryCode, AuthorityInstanceId, AuthorityRole and AuthorityName so that PersonName and Email can be omitted.

Data products for reporters will present the records with active AuthorityStatus and the latest ReportingTime for each unique combination of CountryCode, AuthorityInstanceId, AuthorityRole and Email, including PersonName.

EEA extensions

Authority table will be extended by EEA by several attributes:

Country (for convenience), based on ISO2 country codes,

ReportingTime, registering time of the latest reporting or modification for each record.

Data management rules

Records in Authority table which have AuthorityStatus as inactive for 3 consecutive years will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
AUT_01	CountryCode	varchar(2)	string	PK	countries	
AUT_02	AuthorityInstanceId	varchar(50)	string	PK		
AUT_03	AuthorityRole	varchar(50)	string	PK	authorityrole	
AUT_04	Email	varchar(50)	string	PK		
AUT_05	AuthorityInstance	varchar(20)	string		authorityinstance	
AUT_06	AuthorityName	varchar(150)	string			
AUT_07	AuthorityURL	varchar(150)	string			
AUT_08	AuthorityAddress	varchar(150)	string			
AUT_09	PersonName	varchar(150)	string			
AUT_10	AuthorityStatus	varchar(10)	string		authoritystatus	

Attribute details

AUT_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

AUT_02 – AuthorityInstanceId

Content

Unique identifier depending on authority instance, e.g. ZoneId or NUTS code.

Remarks

If AuthorityInstance is zone then AuthorityInstanceId is a ZoneId, if nuts then nuts code, if network - network id, etc.

It will be cross-checked against ARZ, STA, SPO or MOD table depending on the instance level.

In Reference

View reference attribute

AUT_03 – AuthorityRole

Content

A general object identifier or classification (topic e.g. reporting, assessment etc.).

In Reference

View reference attribute

AUT_04 – Email

Content

Email address of the contact person.

Remarks

Email have to be filled in but reporting details on person's email are necessary only for reporters who need access to the ReportNet3.

An informed consent statement will be attached clarifying that this data will be used for establishing access rights and filling it in means both access request and consent for data use.

In other cases the attribute should be filled in as "Not Reported" or with organisation generic email address.

In Reference

View reference attribute

AUT_05 – AuthorityInstance

Content

Name of the authority instance: zone, network, nuts0, nuts1, nuts2, nuts3, station, SPO, etc.

In Reference

View reference attribute

AUT_06 – AuthorityName

Content

Name of the institute or organization.

In Reference

View reference attribute

AUT_07 – AuthorityURL

Content

Website URL of the institute.

In Reference

View reference attribute

AUT_08 – AuthorityAddress

Content

Address of the institute.

In Reference

View reference attribute

AUT_09 – PersonName

Content

Name of the contact person in the authority or institute.

Remarks

PersonName has to be filled in but reporting details on person's name are necessary only for reporters who need access to the ReportNet3.

An informed consent statement will be attached clarifying that this data will be used for establishing access rights and filling it in means both access request and consent for data use.

In other cases the attribute should be filled in as "Not Reported".

In Reference

View reference attribute

AUT_10 – AuthorityStatus

Content

Status or classification of the authority (e.g. active/inactive).

In Reference

View reference attribute

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
AUT_01	CountryCode	DU	varchar(2)	string	✓	✓
AUT_02	AuthorityInstanceId	DU	varchar(50)	string	✓	
AUT_03	AuthorityRole	1	varchar(50)	string	✓	✓
AUT_04	Email	eve.bot@dema.dus	varchar(50)	string	✓	
AUT_05	AuthorityInstance	nuts0	varchar(20)	string		✓
AUT_06	AuthorityName	Dustovia Environmental Agency (DEA) – Air Quality Division	varchar(150)	string		
AUT_07	AuthorityURL	www.dea.dustovia.dus	varchar(150)	string		
AUT_08	AuthorityAddress	42 CleanAir St, Terminalgrad, Dustovia (DUS-4242)	varchar(150)	string		

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
AUT_09	PersonName	E.V.E. Bot	varchar(150)	string		
AUT_10	AuthorityStatus	active	varchar(10)	string		✓

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

MeasurementStation



Description

EEA extensions

Data management rules

Attributes

Attribute details

STA_01 – CountryCode

STA_02 – StationEoICode

STA_03 – NetworkId

STA_04 – NetworkName

STA_05 – NetworkOrganisationalLevel

STA_06 – Timezone

STA_07 – StationNationalCode

STA_08 – StationName

STA_09 – NetworkDocumentId

Example

Description

The purpose of the MeasurementStation table is to collect data on air quality measuring stations. It includes their EoI codes, names, as well as air quality networks they belong to and time zone in which the networks operate.

MeasurementStation table has to be reported whenever new information or an update of existing information is available. Attributes which create unique identifier of each record: CountryCode and StationEoICode.

Updates of the MeasurementStation table will be managed using reference/legacy data, where each addition and modification will be stamped with ReportingTime:

addition of new record is understood as addition of new combination of values for the attributes creating unique identifier,

modification of existing record is understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC rules to be established, e.g. testing validity of EoI codes.

Valid modifications of existing record in this table are minor corrections, e.g. of NetworkName, StationNationalCode, StationName or Timezone, which are not considered as station re-location into new conditions. Validity of the modifications of existing record will be tested using QC rules to be established. Allowed modifications are not supposed to affect outcomes of compliance analysis and/or assessments.

Data products will present the records with the latest ReportingTime for each unique combination of CountryCode and StationEoICode.

EEA extensions

MeasurementStation table will be extended by EEA by several attributes:

Country (for convenience), based on ISO2 country codes,

City and CityCode, based on intersection of station location with city geometry,

ReportingTime, registering time of the latest reporting or modification for each record.

Data management rules

For traceability purposes it is not allowed to change StationEoICode even in case of significant relocation.

It is recommended not to change NetworkDocumentId in case of documentation update related to the same station/network. Instead, a new attachment should be uploaded in Documentation table re-using the same DocumentId. Previous versions of documentation will be kept in reference data. In case where the same NetworkDocumentId was used for all stations and there is a need to modify

documentation only for one specific station, it is recommended to create a new NetworkDocumentId for that record in MeasurementStation table. Then report old network-specific and new station-specific documents as attachments for that new NetworkDocumentId in the Documentation table in a subsequent manner.

Station without active sampling points will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
STA_01	CountryCode	varchar(2)	string	PK	countries	
STA_02	StationEoICode	varchar(10)	string	PK		SamplingPoint
STA_03	NetworkId	varchar(50)	string			
STA_04	NetworkName	varchar(150)	string			
STA_05	NetworkOrganisationalLevel	varchar(20)	string		administrativelevel	
STA_06	Timezone	varchar(20)	string		timezone	
STA_07	StationNationalCode	varchar(50)	string			
STA_08	StationName	varchar(50)	string			
STA_09	NetworkDocumentId	varchar(150)	string			Documentation

Attribute details

STA_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

STA_02 – StationEoICode

Content

EoI (Exchange of Information) code of the air quality measurement station, as in AirBase, either re-used or given by data provider following strict rules.

Code list / reference

There will be no specific code list in Data Dictionary but the MeasurementStation table in reference data flow will serve as code list for StationEoICode values.

Remarks

StationEoICode must be always provided and cannot be modified.

Station will become inactive if all sampling points in that station are inactive.

The same Station can become active again if there are new active sampling points or existing sampling points become active again.

In Reference

View reference attribute

STA_03 – NetworkId

Content

Identifier of the air quality network, given by data provider.

In Reference

View reference attribute

STA_04 – NetworkName

Content

Name of the air quality measurement network, given by data provider.

In Reference

View reference attribute

STA_05 – NetworkOrganisationalLevel

Content

Level of administration or organization responsible for the air quality network (national, regional, local).

In Reference

View reference attribute

STA_06 – Timezone

Content

Time zone in which air quality measurements and statistics are recorded.

Remarks

TimeZone must be provided and will be used for calculations of AQ statistics.

If TimeZone is modified, all statistics will be recalculated accordingly.

In Reference

View reference attribute

STA_07 – StationNationalCode

Content

National code of the air quality measurement station, given by data provider.

In Reference

View reference attribute

STA_08 – StationName

Content

Name of the air quality measurement station.

In Reference

View reference attribute

STA_09 – NetworkDocumentId

Content

Identifier of the documentation on network and station.

In Reference

View reference attribute

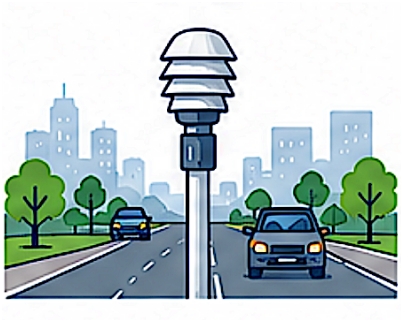
Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
STA_01	CountryCode	DU	varchar(2)	string	✓	✓
STA_02	StationEoICode	DU0001	varchar(10)	string	✓	
STA_03	NetworkId	NET_DU01	varchar(50)	string		
STA_04	NetworkName	Dustovia Capital	varchar(150)	string		
STA_05	NetworkOrganisationalLevel	regional	varchar(20)	string		✓
STA_06	Timezone	UTC-01	varchar(20)	string		✓
STA_07	StationNationalCode	DU01_01	varchar(50)	string		
STA_08	StationName	Rivny-Suburb-A	varchar(50)	string		
STA_09	NetworkDocumentId	DOC_NET_DU01	varchar(150)	string		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

SamplingPoint



Description

EEA extensions

Data management rules

Attributes

Attribute details

SPO_01 – CountryCode

SPO_02 – AssessmentMethodId

SPO_03 – SamplingPointReferenceId

SPO_04 – PollutantId

SPO_05 – StationEoICode

Example

Description

The purpose of the SamplingPoint table is to ensure the link between the AssessmentMethodId, the SamplingPointReferenceId and the StationEoiCode.

Attributes which create unique identifier of each record: CountryCode and AssessmentMethodId.

Updates of the SamplingPoint table will be managed using reference/legacy data, where each addition and modification will be stamped with ReportingTime:

addition of new record is understood as addition of new combination of values for the attributes creating unique identifier,

modification of existing record is understood as modification of a value for any other attribute.

Validity of the new record and modifications will be tested using QC (rules to be established) - e.g. cross checking against existing records in Station table. Allowed modifications are not supposed to affect outcomes of compliance analysis and/or assessments.

EEA extensions

SamplingPoint table will be extended by EEA by several attributes:

Country (for convenience), based on ISO2 country codes,

Pollutant (for convenience), based on code list,

SamplingPointStatus (coming from SamplingPointLocation table), based on values in SamplingProcess.ProcessActivityEnd (if all ProcessId have ProcessActivityEnd populated the SamplingPointStatus is ‘inactive’, otherwise it is ‘active’),

ReportingTime, registering time of the latest reporting or modification for each record.

Data management rules

At the time of creating the first version of reference data, the EEA will establish list of values of SamplingPointReferenceId corresponding to AssessmentMethodIds using rule:

```
'SPO' + EoI code + pollutant code + ordering index
```

(see Identifiers section).

From that moment, countries should generate the SamplingPointReferenceId for all new sampling points/AssessmentMethodId.

SamplingPointReferenceId cannot be modified and cannot be re-used (e.g. after deletion of a sampling point).

SamplingPointReferenceId serves as reference/mapping identifier for situations when a country or a network changes the naming convention for all its measurements (sampling point identifiers); in such cases, the new identifiers (AssessmentMethodIds) should be mapped by reporting authorities to SamplingPointReferenceId corresponding to the same sampling points. The reference data flow will keep both the old and the new identifiers but it will be able to maintain and present continuity of measurement configurations. Situation of changes in naming convention is different from cases when sampling point needs to be closed due to its relocation or changes in air quality conditions (e.g. city sprawl). In such cases all processes within the sampling point should be ended and no new

measurement results should be reported for that AssessmentMethodId. The sampling point will be set as ‘inactive’ in the reference data flow and this will apply as well to the SamplingPointReferenceId. The reporting authority should then create a new sampling point with a new SamplingPointReferenceId.

It is up to the reporting authority to decide when the changes in surrounding conditions are significant enough to inactivate and create sampling points.

SamplingPoint without both active SamplingProcess active SamplingPointLocation and without any ObservationMeasurementResult records with valid values will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SPO_01	CountryCode	varchar(2)	string	PK	countries	
SPO_02	AssessmentMethodId	varchar(100)	string	PK		ComplianceAssessmentMethod SamplingProcess SamplingPointLocation ObservationMeasurementResult
SPO_03	SamplingPointReferenceId	varchar(32)	string			
SPO_04	PollutantId	int	numeric		pollutant	
SPO_05	StationEoICode	varchar(10)	string			MeasurementStation

Attribute details

SPO_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

SPO_02 – AssessmentMethodId

Content

Identifier of the assessment method (sampling point), given by data provider.

Remarks

A sampling point (AssessmentMethodId) can be closed (set as inactive) by ending the ProcessId (specifying ProcessActivityEnd).

The same sampling point (AssessmentMethodId) can be re-open (also with the same ProcessId) by adding new record with new ProcessActivityBegin.

In Reference

View reference attribute

SPO_03 – SamplingPointReferenceId

Content

Reference identifier of the assessment method (sampling point), either re-used or given by data provider following strict rules.

Remarks

Following rules of SPO reference code, unique within CountryCode reported and it can serve as reference code list.

In Reference

View reference attribute

SPO_04 – PollutantId

Content

Code of the air pollutant being measured, as per Data Dictionary standards.

Remarks

Only one for the same AssessmentMethodId/SamplingPointReferenceId.

In Reference

View reference attribute

SPO_05 – StationEoICode

Content

EoI (Exchange of Information) code of the air quality measurement station, as in AirBase, either re-used or given by data provider following strict rules.

Code list / reference

There will be no specific code list in Data Dictionary but the MeasurementStation table in reference data flow will serve as code list for StationEoICode values.

Remarks

StationEoICode must be always provided and cannot be modified.

Station will become inactive if all sampling points in that station are inactive.

The same Station can become active again if there are new active sampling points or existing sampling points become active again.

It will be cross-checked against MeasurementStation table.

In Reference

View reference attribute

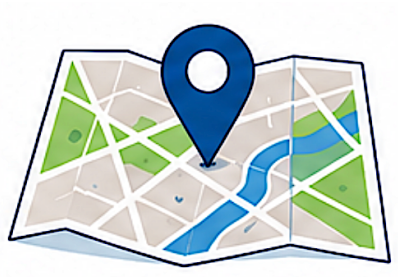
Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SPO_01	CountryCode	DU	varchar(2)	string	✓	✓
SPO_02	AssessmentMethodId	SPO_DU0001_0005_100	varchar(100)	string	✓	
SPO_03	SamplingPointReferenceId	SPOref_DU0001_0005_1	varchar(32)	string		
SPO_04	PollutantId	5	int	numeric		✓
SPO_05	StationEoICode	DU0001	varchar(10)	string		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

SamplingPointLocation



Description

EEA extensions

Data management rules

Attributes

Attribute details

SPL_01 – CountryCode

SPL_02 – AssessmentMethodId

SPL_03 – LocationBegin

SPL_04 – LocationEnd

SPL_05 – StationArea

SPL_06 – SamplingPointCategory

SPL_07 – Hotspot

SPL_08 – Supersite

SPL_09 – Latitude

SPL_10 – Longitude

SPL_11 – Altitude

SPL_12 – InletHeight

SPL_13 – BuildingDistance

SPL_14 – KerbDistance

SPL_15 – EmissionSourceDistance

Example

Description

The purpose of the SamplingPoint table is to collect data on air quality sampling locations. It describes their identifiers, coordinates and other area and location characteristics. The table includes corresponding times of location characteristic effectiveness.

SamplingPointLocation table has to be reported whenever new information or an update of existing information is available. Attributes which create unique identifier of each record: CountryCode, AssessmentMethodId and LocationBegin.

Updates of the SamplingLocation table will not be time stamped, i.e. each update generates the only version of the table in reference/legacy data:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute, e.g. entering end date.

Validity of the new record will be tested using QC rules to be established, e.g. cross-checking relation between the code listed values.

Valid modifications of existing record in this table are minor corrections, e.g. of building distance or coordinates, or altitude, which are not considered as SamplingPoint re-location into new conditions. Validity of the modifications of existing record will be tested using QC rules to be established. Allowed modifications are not supposed to affect outcomes of compliance analysis and/or assessments.

Data products will present all records related to SamplingPoints reported recently for the UTD purpose.

EEA extensions

SamplingPointLocation table will be extended by EEA by several attributes:

Country, for convenience, based on ISO2 country codes,

X and Y, based on coordinates re-projected to SRID3035,

grid number identifiers, based on X and Y in SRID3035,

SamplingPointType, following the previous version of sampling point type classification (traffic, background,industrial); the new type SamplingPointCategory for closed/inactive sampling points will be mapped to the older types using road traffic, background and industry categories, unless countries re-submit meta-data for these sampling points and set their own values; countries should set their own values of SamplingPointCategory for all active sampling points.

Data management rules

In case where the LocationBegin was reported wrongly and/or the corresponding record needs to be deleted, the same value should be reported for LocationEnd. Records where LocationBegin = LocationEnd will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SPL_01	CountryCode	varchar(2)	string	PK	countries	
SPL_02	AssessmentMethodId	varchar(100)	string	PK		ComplianceAssessmentMethod SamplingPoint SamplingProcess ObservationMeasurementResult
SPL_03	LocationBegin	datetime	datetime	PK		
SPL_04	LocationEnd	datetime	datetime			
SPL_05	StationArea	varchar(100)	string		areaclassification	
SPL_06	SamplingPointCategory	varchar(50)	string		samplingpointcategory	
SPL_07	Hotspot	bit	boolean			
SPL_08	Supersite	bit	boolean			
SPL_09	Latitude	decimal(8,4)	numeric			
SPL_10	Longitude	decimal(8,4)	numeric			
SPL_11	Altitude	decimal(10,1)	numeric			
SPL_12	InletHeight	decimal(10,1)	numeric			
SPL_13	BuildingDistance	decimal(10,1)	numeric			
SPL_14	KerbDistance	decimal(10,1)	numeric			

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SPL_15	EmissionSourceDistance	decimal(10,1)	numeric			

Note

Attributes with ReportNet3 data type `date` or `datetime` shall use the ISO 8601 format. Date-time values may include a local UTC offset.

Attribute details

SPL_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

SPL_02 – AssessmentMethodId

Content

Identifier of the assessment method (sampling point), given by data provider.

In Reference

View reference attribute

SPL_03 – LocationBegin

Content

Start time of the location characteristic(s).

Remarks

The permissible modifications in the location characteristics will be verified by adequate QC.

In Reference

View reference attribute

SPL_04 – LocationEnd

Content

End time of the location characteristic(s).

In Reference

View reference attribute

SPL_05 – StationArea

Content

Classification of the air quality measurement station’s area (urban, suburban, rural, etc.).

Remarks

A change in station area implies closing the associated sampling points by ending the SamplingProcess, declaring a new station and the associated new sampling points.

In Reference

View reference attribute

SPL_06 – SamplingPointCategory

Content

Intention or reason for placing sampling point.

Code list / reference

Code list to be developed.

Remarks

Potential categories: traffic, background, industrial, port, airport, residential heating, multisource.

In Reference

View reference attribute

SPL_07 – Hotspot

Content

Indicator if the measurement site is considered as a hotspot.

Remarks

Y/N.

Hotspot relates to the provisions of the recasted Air Quality Directive 2024/2881, Article 4(27) and others.

In Reference

View reference attribute

SPL_08 – Supersite

Content

Indicator if the measurement site is classified as a “Super Site” for advanced monitoring.

Remarks

Y/N.

In Reference

View reference attribute

SPL_09 – Latitude

Content

Latitude coordinate of the air quality station location (decimal degrees).

In Reference

View reference attribute

SPL_10 – Longitude

Content

Longitude coordinate of the air quality station location (decimal degrees).

In Reference

View reference attribute

SPL_11 – Altitude

Content

Altitude of the air quality station location in meters above sea level.

In Reference

View reference attribute

SPL_12 – InletHeight

Content

Height of the sampling inlet where air quality measurements are taken, in meters.

In Reference

View reference attribute

SPL_13 – BuildingDistance

Content

Horizontal distance from the sampling inlet to the nearest building, in meters.

In Reference

View reference attribute

SPL_14 – KerbDistance

Content

Horizontal distance from the sampling inlet to the nearest road kerb, in meters.

In Reference

View reference attribute

SPL_15 – EmissionSourceDistance

Content

Horizontal distance from the main emission source, in meters.

In Reference

View reference attribute

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SPL_01	CountryCode	DU	varchar(2)	string	✓	
SPL_02	AssessmentMethodId	SPO_DU0001_0005_100	varchar(100)	string	✓	
SPL_03	LocationBegin	2013-06-25T00:00	datetime	datetime	✓	
SPL_04	LocationEnd	null	datetime	datetime		
SPL_05	StationArea	urban	varchar(100)	string		✓
SPL_06	SamplingPointCategory	background	varchar(50)	string		✓
SPL_07	Hotspot	N	bit	boolean		
SPL_08	Supersite	N	bit	boolean		
SPL_09	Latitude	43.60625	decimal(8,4)	numeric		
SPL_10	Longitude	33.093242	decimal(8,4)	numeric		
SPL_11	Altitude	21	decimal(10,1)	numeric		
SPL_12	InletHeight	2	decimal(10,1)	numeric		
SPL_13	BuildingDistance	10	decimal(10,1)	numeric		
SPL_14	KerbDistance	5	decimal(10,1)	numeric		

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SPL_15	EmissionSourceDistance	5	decimal(10,1)	numeric		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

SamplingProcess



Description

EEA extensions

Data management rules

Attributes

Attribute details

SPP_01 – CountryCode

SPP_02 – ProcessId

SPP_03 – AssessmentMethodId

SPP_04 – ProcessActivityBegin

SPP_05 – ProcessActivityEnd

SPP_06 – PollutantId

SPP_07 – MeasurementType

SPP_08 – Method

SPP_09 – Equipment

SPP_10 – AnalyticalTechnique

SPP_11 – EquivalenceDemonstrated

SPP_12 – DataQualityDocumentId

SPP_13 – EquivalenceDemonstrationDocumentId

SPP_14 – ProcessDocumentId

Example

Description

The purpose of the SamplingProcess table is to collect data on measurement techniques and methodologies. It describes the specific configurations of equipment, mostly through code listed values, and includes information on quality of the measurements as well as further details about the methods through attached documentation. The table includes corresponding times of operation.

SamplingProcess table has to be reported whenever new information or an update of existing information is available. Attributes which create unique identifier of each record: CountryCode, ProcessId, AssessmentMethodId and ProcessActivityBegin.

Updates of the SamplingProcess table will not be time stamped, i.e. each update generates the only version of the table in reference/legacy data:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute, e.g. entering end date.

Validity of the new record will be tested using QC rules to be established, e.g. cross-checking relation between the code listed values.

Valid modifications of existing record in this table are:

inserting ProcessActivityEnd,

attaching new/different documentation,

minor corrections, e.g. of wrong code list values.

Validity of the modifications of existing record will be tested using QC rules to be established. Allowed modifications are not supposed to affect outcomes of compliance analysis and/or assessments.

Data products will present all records related to SamplingPoints reported recently for the UTD purpose.

EEA extensions

SamplingProcess table will be extended by EEA by several attributes:

Country, for convenience, based on ISO2 country codes,

Pollutant, for convenience, based on code list,

SamplingPointReferenceId, based on SamplingPoint table.

Data management rules

In case where the ProcessActivityBegin was reported wrongly and/or the corresponding record needs to be deleted, the same value should be reported for ProcessActivityEnd. Records where ProcessActivityBegin = ProcessActivityEnd will be flagged for deletion in the reference data flow.

It is recommended not to change DataQualityDocumentId, EquivalenceDemonstrationDocumentId or ProcessDocumentId in case of documentation update related to the same process. Instead, a new attachment should be uploaded in Document table re-using the same DocumentId. Previous versions of documentation will be kept in reference data. In case where the same, e.g. DataQualityDocumentId, was used for all records and there is a need to modify documentation only for one specific process, it is recommended to create a new DataQualityDocumentId for that record in the SamplingProcess table, and then report old non-process-specific and new process-specific documents as attachments for that new DataQualityDocumentId as DocumentId in the Document table in a subsequent manner.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SPP_01	CountryCode	varchar(2)	string	PK	countries	
SPP_02	ProcessId	varchar(150)	string	PK		
SPP_03	AssessmentMethodId	varchar(100)	string	PK		ComplianceAssessmentMethod SamplingPoint SamplingPointLocation ObservationMeasurementResult
SPP_04	ProcessActivityBegin	datetime	datetime	PK		
SPP_05	ProcessActivityEnd	datetime	datetime			
SPP_06	PollutantId	int	numeric		pollutant	
SPP_07	MeasurementType	varchar(50)	string		measurementtype	
SPP_08	Method	varchar(50)	string		measurementmethod	
SPP_09	Equipment	varchar(50)	string		measurementequipment	
SPP_10	AnalyticalTechnique	varchar(50)	string		analyticaltechnique	

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SPP_11	EquivalenceDemonstrated	varchar(50)	string		equivalencedemonstrated	
SPP_12	DataQualityDocumentId	varchar(150)	string			Documentation
SPP_13	EquivalenceDemonstrationDocumentId	varchar(150)	string			Documentation
SPP_14	ProcessDocumentId	varchar(150)	string			Documentation

Note

Attributes with ReportNet3 data type `date` or `datetime` shall use the ISO 8601 format. Date-time values may include a local UTC offset.

Attribute details

SPP_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

SPP_02 – ProcessId

Content

Identifier of the sampling process, given by data provider.

Remarks

The same ProcessId can be re-used for the same equipment configurations under different sampling points (AssessmentMethodId).

In Reference

View reference attribute

SPP_03 – AssessmentMethodId

Content

Identifier of the assessment method (sampling point) given by data provider.

Remarks

A sampling point (AssessmentMethodId) can be closed (set as inactive) by ending the ProcessId (specifying ProcessActivityEnd). The same sampling point can be re-opened, also with the same ProcessId, by adding new record with new ProcessActivityBegin.

In Reference

View reference attribute

SPP_04 – ProcessActivityBegin

Content

Start time of the measurement process.

Remarks

If there is more than one ProcessId within the same AssessmentMethodId, then ProcessActivityBegin - ProcessActivityEnd should not overlap within the same AssessmentMethodId.

In Reference

View reference attribute

SPP_05 – ProcessActivityEnd

Content

End time of the measurement process.

In Reference

View reference attribute

SPP_06 – PollutantId

Content

Code of the air pollutant being measured, as per Data Dictionary standards.

In Reference

View reference attribute

SPP_07 – MeasurementType

Content

Classification of measurement methods into generic types.

In Reference

View reference attribute

SPP_08 – Method

Content

Specific method used for measuring air pollutants.

In Reference

View reference attribute

SPP_09 – Equipment

Content

Equipment used for air pollutant measurement.

In Reference

View reference attribute

SPP_10 – AnalyticalTechnique

Content

Analytical technique used for measuring pollutants.

In Reference

View reference attribute

SPP_11 – EquivalenceDemonstrated

Content

Status of equivalence demonstration according to regulatory requirements.

In Reference

View reference attribute

SPP_12 – DataQualityDocumentId

Content

Identifier of the Quality Assurance report given by data provider.

In Reference

View reference attribute

SPP_13 – EquivalenceDemonstrationDocumentId

Content

Identifier of the Equivalence demonstration report given by data provider.

In Reference

View reference attribute

SPP_14 – ProcessDocumentId

Content

Identifier of the documentation on process and data quality given by data provider.

In Reference

View reference attribute

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SPP_01	CountryCode	DU	varchar(2)	string	✓	✓
SPP_02	ProcessId	SPP_DU0001_0005_1	varchar(150)	string	✓	
SPP_03	AssessmentMethodId	SPO_DU0001_0005_100	varchar(100)	string	✓	
SPP_04	ProcessActivityBegin	2013-06-25T00:00	datetime	datetime	✓	
SPP_05	ProcessActivityEnd	null	datetime	datetime		
SPP_06	PollutantId	5	int	numeric		✓
SPP_07	MeasurementType	automatic	varchar(50)	string		✓
SPP_08	Method	BETA	varchar(50)	string		✓
SPP_09	Equipment	BETA1020	varchar(50)	string		✓
SPP_10	AnalyticalTechnique	null	varchar(50)	string		✓
SPP_11	EquivalenceDemonstrated	yes	varchar(50)	string		✓
SPP_12	DataQualityDocumentId	DOC_DQ_SPP_DU0005_1	varchar(150)	string		
SPP_13	EquivalenceDemonstrationDocumentId	DOC_EQ_SPP_DU0005_1	varchar(150)	string		
SPP_14	ProcessDocumentId	DOC_PR_SPP_DU0005_1	varchar(150)	string		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

ModelObjectiveEstimation



Description

EEA extensions

Data management rules

Attributes

Attribute details

MOE_01 – CountryCode

MOE_02 – AssessmentMethodId

MOE_03 – DataAggregationProcessId

MOE_04 – AssessmentMethodName

MOE_05 – PollutantId

MOE_06 – ResultEncoding

MOE_07 – MethodApplication

MOE_08 – GenericMQI

MOE_09 – DataQualityDocumentId

MOE_10 – MethodDocumentId

Example

Description

The purpose of the ModelObjectiveEstimation table is to collect data on applications of modelling (MOD) or objective estimation (OBE) methodology. It describes the specific use of the model/OBE (assessments, adjustments, scenarios) and the way the modelling/OBE results are provided and should be interpreted (via ResultEncoding). It also includes information on MQI and the methodology itself through attached documentation.

ModelObjectiveEstimation (record) table has to be reported whenever a new information or an update of existing information is available. Attributes which create unique identifier of each record: CountryCode, AssessmentMethodId and DataAggregationProcessId.

Updates of the ModelObjectiveEstimation table will not be time stamped - i.e. each update generates the only version of the table in reference/legacy data:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against code lists.

Valid modifications of existing record in this table are:

attaching new/different documentation,

minor corrections e.g. of wrong code list values or MQI value.

Validity of the modifications of existing record will be tested using QC (rules to be established). Allowed modifications are not supposed to affect outcomes of compliance analysis and/or assessments.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported recently, highlighting the differences between validated reports and recent reports in case of re-submissions.

EEA extensions

ModelObjectiveEstimation table will be extended by EEA by several attributes:

Country (for convenience), based on ISO2 country codes,

Pollutant (for convenience), based on code list,

DataAggregationProcess, based on DataAggregationProcessId.

Data management rules

The ModelObjectiveEstimation table records may be kept relatively static, i.e. there is no need to report a new AssessmentMethodId every year if the model/OBE application does not change significantly.

Details regarding annual quality indicators or input data can be provided with new documentation attached to the same document Ids (see more clarification below). It is up to the reporting authorities to decide if the reporting of data from models require new entry in the ModelObjectiveEstimation table. Good practice is to check if it is possible to report MOEResult (Inline or External) with the same AssessmentMethodId without getting a BLOCKER error - usually if this works, then it is fine to keep the same record in the ModelObjectiveEstimation table.

The GenericMQI should be understood as MQI calculated for the reported model/OBE but not specific to the related results - e.g. MQI for models/OBE used for reporting adjustment (Saharan dust, etc.) or for reporting SRs (MQI would not necessarily reflect the uncertainty of the SR areas), or for reporting scenarios.

For the purpose of reporting a single value for air quality zone (usually from OBE), ResultEncoding = ‘zone’ should be used. Since the value is already reported in ComplianceAssessmentMethod table, no corresponding records in MOEResultInline neither in MOEResultExternal will be expected in such case.

It is recommended not to change ModelDocumentId or DataQualityDocumentId in case of documentation update related to the same modelling/OBE application. Instead, a new attachment should be uploaded in Document table re-using the same DocumentId. Previous versions of documentation will be kept in reference data. In case when the same e.g. DataQualityDocumentId was used for all records and there is a need to modify documentation only for one specific model/OBE, it is recommended to create a new DataQualityDocumentId for that record in ModelObjectiveEstimation table, and then report old (not model/OBE specific) and new (model/OBE specific) documents as attachments for that new DataQualityDocumentId (as DocumentId) in the Document table in a subsequent manner.

Records in ModelObjectiveEstimation table that do not have references in other tables and do not have any valid values in MOEResultInline nor in MOEResultExternal will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
MOE_01	CountryCode	varchar(2)	string	PK	countries	

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
MOE_02	AssessmentMethodId	varchar(100)	string	PK		ComplianceAssessmentMethod SpatialRepresentativeness PollutionLevelAdjustment PlanScenario MOEResultInline MOEResultExternal
MOE_03	DataAggregationProcessId	varchar(50)	string	PK	aggregationprocess	MOEResultInline MOEResultExternal
MOE_04	AssessmentMethodName	varchar(150)	string			
MOE_05	PollutantId	int	numeric		pollutant	
MOE_06	ResultEncoding	varchar(10)	string		resultencoding	
MOE_07	MethodApplication	varchar(20)	string		modelapplication	
MOE_08	GenericMQI	decimal(5,2)	numeric			
MOE_09	DataQualityDocumentId	varchar(150)	string			Documentation
MOE_10	MethodDocumentId	varchar(150)	string			Documentation

Attribute details

MOE_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

MOE_02 – AssessmentMethodId

Content

Identifier of the assessment method (model), given by data provider.

Remarks

AssessmentMethodId which is model or OBE identifier is understood as an application of modelling or estimation methodology. It may be re-used for different years if the methodology and data sources remain the same (for example when the same source of meteorological data is used with data from

different years). This attribute must follow a defined structure (see Identifiers section).

In Reference

View reference attribute

MOE_03 – DataAggregationProcessId

Content

Identifier for the process of data aggregation into statistical values.

Remarks

Several different statistics may be generated by the same AssessmentMethodId (for example where time series are modelled and subsequently aggregated).

In Reference

View reference attribute

MOE_04 – AssessmentMethodName

Content

Name of model or OBE which can be re-used with different AssessmentMethodId values.

Remarks

Examples include model names such as ‘CHIMERE’, ‘WRF-Chem’, etc.

In Reference

View reference attribute

MOE_05 – PollutantId

Content

Code of the air pollutant being measured, as per Data Dictionary standards.

Remarks

Only one pollutant can be handled by a single AssessmentMethodId (model/OBE).

In Reference

View reference attribute

MOE_06 – ResultEncoding

Content

Encoding method used for model results.

Remarks

Allowed values include:

zone

inline

external

In Reference

View reference attribute

MOE_07 – MethodApplication

Content

Purpose of the model application.

Examples include:

assessment

adjustment

scenario

spatial representativeness

Remarks

The same Model/OBE identifier can only have one MethodApplication.

In Reference

View reference attribute

MOE_08 – GenericMQI

Content

Modelling Quality Indicator.

Remarks

MQI is reported here because it may originate from model or OBE validation exercises not directly related to the area for which modelling results are submitted within AQ e-Reporting. It is considered general information describing the quality of the modelling application used in the reporting.

In Reference

View reference attribute

MOE_09 – DataQualityDocumentId

Content

Identifier of the report describing the model, given by the data provider.

In Reference

View reference attribute

MOE_10 – MethodDocumentId

Content

Identifier of the report detailing the quality of the modelled data, given by the data provider.

In Reference

View reference attribute

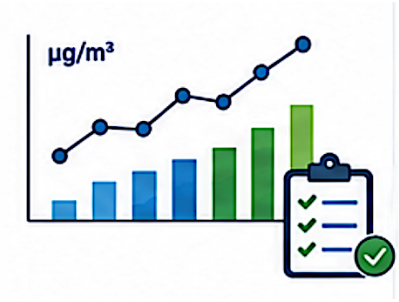
Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
MOE_01	CountryCode	DU	varchar(2)	string	✓	✓
MOE_02	AssessmentMethodId	MOD_DU_NO2	varchar(100)	string	✓	
MOE_03	DataAggregationProcessId	P1Y	varchar(50)	string	✓	✓
MOE_04	AssessmentMethodName	null	varchar(150)	string		
MOE_05	PollutantId	8	int	numeric		✓
MOE_06	ResultEncoding	external	varchar(10)	string		✓
MOE_07	MethodApplication	assessment	varchar(20)	string		✓
MOE_08	GenericMQI	0.8	decimal(5,2)	numeric		
MOE_09	DataQualityDocumentId	DOC_MOD_DQ_MOD_DU_NO2	varchar(150)	string		
MOE_10	MethodDocumentId	DOC_MOD_MR_MOD_DU_NO2	varchar(150)	string		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

ObservationMeasurementResult



Description

EEA extensions

Data management rules

Attributes

Attribute details

OMR_01 – CountryCode

OMR_02 – AssessmentMethodId

OMR_03 – Start

OMR_04 – PollutantId

OMR_05 – End

OMR_06 – Value

OMR_07 – Unit

OMR_08 – Validity

OMR_09 – Verification

OMR_10 – DataCapture

OMR_11 – TimeResolution

OMR_12 – ResultTime

Example

Description

The purpose of the ObservationMeasurementResult table is to collect information on air quality measurement results. It describes the AQ values and time reference for each value. Location of each measurement can be identified through the SamplingPointLocation table.

ObservationMeasurementResult table has to be reported every year because new information is expected at each reporting cycle. Attributes which create unique identifier of each record: CountryCode, AssessmentMethodId and Start.

Updates of the ObservationMeasurementResult table will not be time stamped - i.e. each update generates the only version of the time series with AQ values that is stored at the EEA, as well as the only version of the AQ statistics stored in the reference/legacy data:

either as addition of a new record, understood as addition of a new combination of values for the attributes creating the unique identifier;

or modification of an existing record, understood as modification of a value for any other attribute.

Validity of new records will be tested using QC procedures (rules to be established), including cross-checking against existing records in the SamplingPoint table.

Valid modifications of existing records are based on re-submission of time series for the existing set of attributes creating the unique identifier:

either to overwrite or correct values of other attributes;

or to modify the Validity flag to `-1` for deletion purposes.

Validity of modifications will be tested through QC procedures. Allowed modifications may affect outcomes of compliance analysis and/or assessments. Therefore, re-submissions for past years, for which compliance reporting has already been completed, will require additional tests against reference/legacy data before official release of such data is granted.

Data products for the public (download services for time series and aggregated values) will present records validated at the latest deadline. Data products for reporters will present records as most recently reported, highlighting differences between validated reports and recent reports in case of re-submissions.

EEA extensions

Reference data flow will contain two tables representing reporting of time series:

ObservationMeasurementInventory;

ObservationMeasurementStatistics.

ObservationMeasurementInventory will collect information about counts of values reported for each sampling point and year with specific settings (validity, verification, time resolution, etc.) and will contain a reference to the raw data file. ObservationMeasurementStatistics in the reference/legacy data flow will differ significantly from the reporting table because it will only contain aggregated values.

The scope of aggregations stored in that table remains to be agreed (P1D ?, P1M ?, annual only ?, etc.). SamplingPointReferenceId will be used in the reference/legacy data flow together with AssessmentMethodId in order to preserve continuity.

ObservationMeasurementStatistics will also be extended by attributes such as:

Pollutant (for convenience), based on code list;

DataCoverage, as a feature of aggregated data;

DataAggregationProcessId (P1Y, etc.);

DataAggregationProcess, based on DataAggregationProcessId;

SourceDataFlow (AirBase, R2 e-Reporting, R3 e-Reporting, UTD).

Data management rules

Deletion of records from ObservationMeasurementResult can be achieved by re-submitting the same records with the Validity flag set to `-1`.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
OMR_01	CountryCode	varchar(2)	string	PK	countries	
OMR_02	AssessmentMethodId	varchar(100)	string	PK		ComplianceAssessmentMethod SamplingPoint SamplingPointLocation SamplingProcess
OMR_03	Start	datetime	datetime	PK		
OMR_04	PollutantId	int	numeric		pollutant	
OMR_05	End	datetime	datetime			

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
OMR_06	Value	decimal(10,2)	numeric			
OMR_07	Unit	varchar(10)	string		concentrationunit	
OMR_08	Validity	int	numeric		observationvalidity	
OMR_09	Verification	int	numeric		observationverification	
OMR_10	DataCapture	decimal(5,2)	numeric			
OMR_11	TimeResolution	varchar(10)	string		primaryObservation	
OMR_12	ResultTime	datetime	datetime			

Note

Attributes with ReportNet3 data type `date` or `datetime` shall use the ISO 8601 format. Date-time values may include a local UTC offset.

Attribute details

OMR_01 – CountryCode

Content

Country or territory ISO2 code.

OMR_02 – AssessmentMethodId

Content

Identifier of the assessment method (sampling point), given by data provider.

Remarks

It will be cross-checked against the SamplingPoint table.

OMR_03 – Start

Content

Start date and time of the measurement period.

Remarks

Start must be provided with time zone information. This information will be used for recalculation of all datetimes into UTC+1 for storage and analytical purposes.

OMR_04 – PollutantId

Content

Code of the air pollutant being measured, as per Data Dictionary standards.

OMR_05 – End

Content

End date and time of the measurement period.

Remarks

End must be provided with time zone information. This information will be used for recalculation of all datetimes into UTC+1 for storage and analytical purposes.

OMR_06 – Value

Content

Measured concentration or level of the air pollutant.

OMR_07 – Unit

Content

Unit of measurement for the air pollution level (e.g. µg/m³).

Remarks

Units must follow those recommended for AQD pollutants.

OMR_08 – Validity

Content

Indicator of whether the measurement data is valid or not.

OMR_09 – Verification

Content

Information based on verification flags found in reported time series.

OMR_10 – DataCapture

Content

The proportion of valid measurement time relative to total measured time, expressed as a percentage.

OMR_11 – TimeResolution

Content

Time resolution of the reported measurement observations (e.g. hourly, daily).

Remarks

TimeResolution must be consistent with End minus Start and will be tested through QC procedures.

OMR_12 – ResultTime

Content

Time at which the result was generated or recorded.

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
OMR_01	CountryCode	DU	varchar(2)	string	✓	✓
OMR_02	AssessmentMethodId	SPO_DU0001_0005_100	varchar(100)	string	✓	
OMR_03	Start	2024-01-01T00:00:00	datetime	datetime	✓	
OMR_04	PollutantId	5	int	numeric		✓
OMR_05	End	2024-01-01T01:00:00	datetime	datetime		
OMR_06	Value	16.4	decimal(10,2)	numeric		
OMR_07	Unit	ug/m3	varchar(10)	string		✓

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
OMR_08	Validity	1	int	numeric		✓
OMR_09	Verification	1	int	numeric		✓
OMR_10	DataCapture		decimal(5,2)	numeric		
OMR_11	TimeResolution	hour	varchar(10)	string		✓
OMR_12	ResultTime	2025-09-28T16:12:36	datetime	datetime		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

MOEResultInline



Description

EEA extensions

Data management rules

Attributes

Attribute details

MRI_01 – CountryCode

MRI_02 – AssessmentMethodId

MRI_03 – Start

MRI_04 – DataAggregationProcessId

MRI_05 – X

MRI_06 – Y

MRI_07 – PollutantId

MRI_08 – End

MRI_09 – Value

MRI_10 – Unit

MRI_11 – Validity

MRI_12 – SpatialResolution

MRI_13 – ResultTime

Example

Description

The purpose of the MOEResultInline table is to collect information on (aggregated and gridded) results from air quality modelling applications. It describes the AQ values, time reference and the area of modelling in a set of grid cells following the common grid approach (see Introduction).

MOEResultInline table has to be reported every year (if models/OBE are declared for compliance) because new information is expected at each reporting cycle. Attributes which create unique identifier of each record: CountryCode, AssessmentMethodId, Start, DataAggregationProcessId, X and Y.

Updates of the MOEResultInline table will not be time stamped - i.e. each update generates the only version of the (gridded) AQ values that is stored at the EEA, as well - the only version of the corresponding table in reference/legacy data:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against existing records in ModelObjectiveEstimation table.

Valid modifications of existing record in this table are based on re-submission of the (gridded) AQ values for the existing set of attributes creating unique identifier:

either to overwrite/correct the values of other attributes,

or to modify Validity flag to -1 for deletion purpose.

Validity of the modifications of existing record will be tested using QC (rules to be established). Allowed modifications may affect outcomes of compliance analysis and/or assessments, therefore re-submissions for past years, for which compliance reporting has been already done, will require additional tests vs reference/legacy data before official release of such data is granted.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported recently, highlighting the differences between validated reports and recent reports in case of re-submissions.

EEA extensions

Reference data flow will have two tables representing reporting of model/OBE results:

MOEResultInventory;

MOEResultGrid.

MOEResultInventory will collect information about count of values reported for each model/OBE and year with certain settings (validity, verification, spatial resolution, etc.) and will contain a reference to raw data file.

MOEResultGrid table will differ from the reporting data flow however it will contain gridded AQ values in 3 layers:

2 Very High Resolution layers (10 and 100m - depending on original resolutions) for populated areas, traffic links and areas with values above thresholds,

1 High Resolution layer (1000m) for all areas.

MOEResultGrid table will be extended by several attributes:

Pollutant (for convenience), based on code list,

DataAggregationProcess, based on DataAggregationProcessId,

SourceDataFlow (R2 e-Reporting, R3 e-Reporting),

Verification,

Grid number identifiers, based on X,Y (SRID 3035).

Data management rules

Results from modelling reported inline should point to record in ModelObjectiveEstimation table with ResultEncoding = ‘ inline’.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
MRI_01	CountryCode	varchar(2)	string	PK	countries	

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
MRI_02	AssessmentMethodId	varchar(100)	string	PK		ComplianceAssessmentMethod ModelObjectiveEstimation
MRI_03	Start	datetime	datetime	PK		
MRI_04	DataAggregationProcessId	varchar(50)	string	PK	aggregationprocess	ModelObjectiveEstimation
MRI_05	X	bigint	numeric	PK		
MRI_06	Y	bigint	numeric	PK		
MRI_07	PollutantId	int	numeric		pollutant	
MRI_08	End	datetime	datetime			
MRI_09	Value	decimal(10,2)	numeric			
MRI_10	Unit	varchar(10)	string		concentrationunits	
MRI_11	Validity	int	numeric		observationvalidity	
MRI_12	SpatialResolution	int	numeric		spatialresolution	
MRI_13	ResultTime	datetime	datetime			

Note

Attributes with ReportNet3 data type `date` or `datetime` shall use the ISO 8601 format. Date-time values may include a local UTC offset.

Attribute details

MRI_01 – CountryCode

Content

Country or territory ISO2 code.

MRI_02 – AssessmentMethodId

Content

Identifier of the assessment method (model), given by provider. This attribute must follow a defined structure (see Identifiers section).

Remarks

It will be cross-checked against the ModelObjectiveEstimation table.

MRI_03 – Start

Content

Start date and time of the model simulation period.

Remarks

Start must be provided with time zone. This information will be used for recalculation of all datetimes into UTC+1 for storage and analytical purposes.

MRI_04 – DataAggregationProcessId

Content

Identifier for the process of aggregating model data into statistical values.

MRI_05 – X

Content

X-coordinate of the modeled data point in a projected coordinate reference system.

Remarks

It must be using SRID 3035.

MRI_06 – Y

Content

Y-coordinate of the modeled data point in a projected coordinate reference system.

Remarks

It must be using SRID 3035.

MRI_07 – PollutantId

Content

Code of the air pollutant being modeled, as per Data Dictionary standards.

MRI_08 – End

Content

End date and time of the model simulation period.

Remarks

End must be provided with time zone. This information will be used for recalculation of all datetimes into UTC+1 for storage and analytical purposes.

MRI_09 – Value

Content

Modeled concentration or level of the air pollutant.

MRI_10 – Unit

Content

Unit of measurement for the air pollution level (e.g., µg/m³).

Remarks

Units must follow those recommended unit for AQD pollutants.

MRI_11 – Validity

Content

Indicator of whether the model data is valid or flagged.

MRI_12 – SpatialResolution

Content

Description of the spatial resolution of the model output, indicating the level of detail.

Remarks

SpatialResolution will be tested against code list for allowed values.

MRI_13 – ResultTime

Content

Time at which the model results were generated or recorded.

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
MRI_01	CountryCode	DU	varchar(2)	string	✓	✓
MRI_02	AssessmentMethodId	OBE_DU_SEASALT_PM10_H_LV_ADJ	varchar(100)	string	✓	
MRI_03	Start	2024-01-01T00:00:00	datetime	datetime	✓	
MRI_04	DataAggregationProcessId	P1D	varchar(50)	string	✓	✓
MRI_05	X	3463000	bigint	numeric	✓	
MRI_06	Y	2245000	bigint	numeric	✓	
MRI_07	PollutantId	5	int	numeric		✓
MRI_08	End	2024-01-02T00:00:00	datetime	datetime		
MRI_09	Value	1	decimal(10,2)	numeric		
MRI_10	Unit	ug/m3	varchar(10)	string		✓
MRI_11	Validity	1	int	numeric		✓
MRI_12	SpatialResolution	1000	int	numeric		✓
MRI_13	ResultTime	2025-09-14T10:19:44	datetime	datetime		

Legend: PK = Primary Key; FK = Foreign Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

MOEResultExternal



Description

EEA extensions

Data management rules

Attributes

Attribute details

MRE_01 – CountryCode

MRE_02 – AssessmentMethodId

MRE_03 – Start

MRE_04 – DataAggregationProcessId

MRE_05 – PollutantId

MRE_06 – End

MRE_07 – Unit

MRE_08 – Validity

MRE_09 – SpatialResolution

MRE_10 – ResultTime

MRE_11 – GeoTiffAttachment

Example

Description

The purpose of the MOEResultExternal table is to collect information on results from air quality modelling applications. It is similar to the table MOEResultInline except that here the results are submitted in attached GEOTIFF file, following the common grid approach (see Introduction).

MOEResultExternal table has to be reported every year (if models/OBE are declared for compliance) because new information is expected at each reporting cycle. Attributes which create unique identifier of each record: CountryCode, AssessmentMethodId, Start and DataAggregationProcessId.

Updates of the MOEResultExternal table will not be time stamped - i.e. each update generates the only version of the (gridded) AQ values that is stored at the EEA, as well - the only version of the corresponding table in reference/legacy data:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against existing records in ModelObjectiveEstimation table.

Valid modifications of existing record in this table are based on re-submission of a GEOTIFF file for the existing set of attributes creating unique identifier:

either to overwrite/correct the values of other attributes,

or to modify Validity flag to -1 for deletion purpose.

Validity of the modifications will be tested using QC (rules to be established). Allowed modifications may affect outcomes of compliance analysis and/or assessments. Therefore, re-submissions for past years, for which compliance reporting has already been completed, will require additional tests against reference/legacy data before official release of such data is granted.

Data products for the public will present records validated at the latest deadline. Data products for reporters will present records as most recently reported, highlighting differences between validated reports and recent reports in case of re-submissions.

EEA extensions

Reference data flow will have two tables representing reporting of model/OBE results:

MOEResultInventory;

MOEResultGrid.

MOEResultInventory will collect information about count of values reported for each model/OBE and year with certain settings (validity, verification, spatial resolution, etc.) and will contain a reference to raw data file. MOEResultGrid table will differ from the reporting data flow however it will contain gridded AQ values in 3 layers:

2 Very High Resolution layers (10 and 100m - depending on original resolutions) for populated areas, traffic links and areas with values above thresholds,

1 High Resolution layer (1000m) for all areas.

MOEResultGrid table will be extended by several attributes:

Pollutant (for convenience), based on code list,

DataAggregationProcess, based on DataAggregationProcessId,

SourceDataFlow (R2 e-Reporting, R3 e-Reporting),

Verification,

Grid number identifiers, based on X,Y (SRID 3035).

Data management rules

Results from modelling reported through external files should point to a record in the ModelObjectiveEstimation table with:

```
ResultEncoding = 'external'
```

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
MRE_01	CountryCode	varchar(2)	string	PK	countries	
MRE_02	AssessmentMethodId	varchar(100)	string	PK		ComplianceAssessmentMethod ModelObjectiveEstimation
MRE_03	Start	datetime	datetime	PK		
MRE_04	DataAggregationProcessId	varchar(50)	string	PK	aggregationprocess	ModelObjectiveEstimation
MRE_05	PollutantId	int	numeric		pollutant	

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
MRE_06	End	datetime	datetime			
MRE_07	Unit	varchar(10)	string		concentrationunits	
MRE_08	Validity	int	numeric		observationvalidity	
MRE_09	SpatialResolution	int	numeric		spatialresolution	
MRE_10	ResultTime	datetime	datetime			
MRE_11	GeoTiffAttachment	varchar(100)	attachment (R3 data type)			

Note

Attributes with ReportNet3 data type `date` or `datetime` shall use the ISO 8601 format. Date-time values may include a local UTC offset.

Attribute details

MRE_01 – CountryCode

Content

Country or territory ISO2 code.

MRE_02 – AssessmentMethodId

Content

Identifier of the assessment method (model), given by provider.

Remarks

It will be cross-checked against the ModelObjectiveEstimation table. This attribute must follow a defined structure (see Identifiers section).

MRE_03 – Start

Content

Start date and time of the model simulation period.

Remarks

Start must be provided with time zone. This information will be used for recalculation of all datetimes into UTC+1 for storage and analytical purposes.

MRE_04 – DataAggregationProcessId

Content

Identifier for the process of aggregating model data into statistical values.

MRE_05 – PollutantId

Content

Code of the air pollutant being modeled, as per Data Dictionary standards.

MRE_06 – End

Content

End date and time of the model simulation period.

Remarks

End must be provided with time zone. This information will be used for recalculation of all datetimes into UTC+1 for storage and analytical purposes.

MRE_07 – Unit

Content

Unit of measurement for the air pollution level (e.g., µg/m³).

Remarks

Units must follow those recommended unit for AQD pollutants.

MRE_08 – Validity

Content

Indicator of whether the model data is valid or flagged.

MRE_09 – SpatialResolution

Content

Description of the spatial resolution of the model output, indicating the level of detail - new code list.

Remarks

SpatialResolution will be tested against the code list for allowed values.

MRE_10 – ResultTime

Content

Time at which the model results were generated or recorded.

MRE_11 – GeoTiffAttachment

Content

Attached GEOTIFF.

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
MRE_01	CountryCode	DU	varchar(2)	string	✓	✓
MRE_02	AssessmentMethodId	MOD_DU_PM10_NAT_SR	varchar(100)	string	✓	
MRE_03	Start	2021-01-01T00:00:00	datetime	datetime	✓	
MRE_04	DataAggregationProcessId	P1Y	varchar(50)	string	✓	✓
MRE_05	PollutantId	5	int	numeric		✓
MRE_06	End	2022-01-01T00:00:00	datetime	datetime		
MRE_07	Unit	ug/m3	varchar(10)	string		✓
MRE_08	Validity	1	int	numeric		✓
MRE_09	SpatialResolution	1000	int	numeric		✓
MRE_10	ResultTime	2023-04-14T10:19:44	datetime	datetime		
MRE_11	GeoTiffAttachment		varchar(100)	attachment (R3 data type)		

Legend: PK = Primary Key; FK = Foreign Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

AssessmentRegimeZone



Description

EEA extensions

Data management rules

Attributes

Attribute details

ARZ_01 – CountryCode

ARZ_02 – AssessmentRegimeId

ARZ_03 – ZoneId

ARZ_04 – ZoneNationalCode

ARZ_05 – ZoneArea

ARZ_06 – ZoneCategory

ARZ_07 – ZoneType

ARZ_08 – ZoneName

ARZ_09 – PollutantId

ARZ_10 – ProtectionTarget

ARZ_11 – ObjectiveType

ARZ_12 – ReportingMetric

ARZ_13 – AssessmentThresholdExceedance

ARZ_14 – PostponementYear

ARZ_15 – FixedMeasurementReduction

ARZ_16 – ZoneResidentPopulationYear

ARZ_17 – ZoneResidentPopulation

ARZ_18 – ClassificationYear

ARZ_19 – ClassificationDocumentId

Example

Description

The purpose of the AssessmentRegimeZone table is to collect information about air quality zones and related assessment regimes. It specifies zone details such as (geographical) name, area, code, category, type, population as well as zone level parameters of assessment regimes such as objective, target, pollutant, reporting metric, assessment threshold and characteristics of exemptions (postponement, SPO reduction).

AssessmentRegimeZone table has to be reported at least every 5 years or whenever a new information or an update of existing information is available. Attributes which create unique identifier of each record: CountryCode and AssessmentRegimeId.

Updates of the AssessmentRegimeZone table will not be time stamped, i.e. each update generates the only version of the assessment regimes that is stored at the EEA:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute. Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against aggregation code list, existing records in Zone table, etc.

Valid modifications of existing record in this table are:

correction of PostponementYear or FixedMeasurementReduction or AssessmentThresholdExceedance,

minor corrections e.g. of ZoneName or ZoneNationalCode, ZoneResidentPopulation and ZoneResidentPopulationYear.

Validity of the modifications of existing record will be tested using QC (rules to be established). Allowed modifications may affect outcomes of compliance analysis and/or assessments, therefore re-submissions for past years, for which compliance reporting has been already done, will require additional tests vs reference/legacy data before official release of such data is granted.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported recently, highlighting the differences between validated reports and recent reports in case of re-submissions.

EEA extensions

AssessmentRegimeZone table will be extended by EEA by several attributes in the reference data flow:

Country (for convenience), based on ISO2 country codes,

Pollutant (for convenience), based on code list,

DataAggregationProcess, based on DataAggregationProcessId,

ReportingYear, based on the corresponding records in ComplianceAssessmentMethod,

Required number of sampling points, based on the recasted Air Quality Directive 2024/2881,

Reported number of sampling points and modelling applications, based on data reported in ComplianceAssessmentMethod table,

SR areas coverage,

Modelling results coverage,

Assessment regime zone exceedance status,

Exceedance extent coverage,

Reported number of measures.

Data management rules

It is recommended to report the AssessmentRegimeZone table only when a new classification is available (at least every 5 years).

If a NUTS region is used - the ZoneCategory should be reported as `nuts` and NUTS region id should be provided as ZoneId, geometry does not need to be reported. If a combination of NUTS regions is used, the area should be reported in the same way as a zone, i.e. ZoneCategory should be reported as `aqzone`, ZoneId should be assigned (following the rule for ZoneIds) and the zone geometry must be provided.

In case of reporting territorial units for AEI, the NUTS regions or combination of NUTS region may be used.

The new environmental objectives should be reported from the beginning of the new reporting in ReportNet3, to cover the needs for roadmaps.

Records in AssessmentRegimeZone table which do not have any correspondence to records in ComplianceAssessmentMethods table will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
ARZ_01	CountryCode	varchar(2)	string	PK	countries	
ARZ_02	AssessmentRegimeId	varchar(50)	string	PK		ComplianceAssessmentMethod
ARZ_03	ZoneId	varchar(50)	string			ZoneGeometry
ARZ_04	ZoneNationalCode	varchar(50)	string			
ARZ_05	ZoneArea	decimal(10,2)	numeric			
ARZ_06	ZoneCategory	varchar(20)	string		zonecategory	
ARZ_07	ZoneType	varchar(20)	string		zonetype	
ARZ_08	ZoneName	varchar(150)	string			
ARZ_09	PollutantId	int	numeric		pollutant	
ARZ_10	ProtectionTarget	varchar(50)	string		protectiontarget	
ARZ_11	ObjectiveType	varchar(50)	string		objectivetype	
ARZ_12	ReportingMetric	varchar(50)	string		reportingmetric	
ARZ_13	AssessmentThresholdExceedance	varchar(20)	string		assessmentthresholdexceedance	
ARZ_14	PostponementYear	int	numeric			
ARZ_15	FixedMeasurementReduction	bit	boolean			

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
ARZ_16	ZoneResidentPopulationYear	int	numeric			
ARZ_17	ZoneResidentPopulation	int	numeric			
ARZ_18	ClassificationYear	int	numeric			
ARZ_19	ClassificationDocumentId	varchar(150)	string			Documentation

Attribute details

ARZ_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

ARZ_02 – AssessmentRegimeId

Content

Identifier of the air quality assessment regime, given by data provider.

Remarks

This attribute must follow a defined structure (see Identifiers section).

In Reference

View reference attribute

ARZ_03 – ZoneId

Content

Identifier of the air quality zone, given by data provider.

Remarks

It will be cross-checked against the ZoneGeometry table.

In Reference

View reference attribute

ARZ_04 – ZoneNationalCode

Content

Unique identifier of the air quality zone, given by data provider.

In Reference

View reference attribute

ARZ_05 – ZoneArea

Content

Total area of the air quality zone.

In Reference

View reference attribute

ARZ_06 – ZoneCategory

Content

Category of the air quality zone (aq zone or nuts) - new code list.

In Reference

View reference attribute

ARZ_07 – ZoneType

Content

Type of air quality zone (e.g., agglomeration and non-agglomeration).

In Reference

View reference attribute

ARZ_08 – ZoneName

Content

Geographical name of the air quality zone.

In Reference

View reference attribute

ARZ_09 – PollutantId

Content

Code of the air pollutant for which the assessment is being conducted.

In Reference

View reference attribute

ARZ_10 – ProtectionTarget

Content

Protection target for air quality (e.g., human health, vegetation).

In Reference

View reference attribute

ARZ_11 – ObjectiveType

Content

Type of environmental objective related to air quality assessment.

In Reference

View reference attribute

ARZ_12 – ReportingMetric

Content

Metric corresponding to the standard (objective type).

In Reference

View reference attribute

ARZ_13 – AssessmentThresholdExceedance

Content

Status indicating whether an assessment threshold has been exceeded.

In Reference

View reference attribute

ARZ_14 – PostponementYear

Content

Year of postponement applied to the air quality zone, pollutant, target, objective and metric/aggregation.

Remarks

PostponementYear relates to the postponement of attainment deadline and exemption from the obligation to apply certain limit values foreseen in the recasted Air Quality Directive 2024/2881 art. 18. Leaving this attribute blank will mean that no postponement does apply.

In Reference

View reference attribute

ARZ_15 – FixedMeasurementReduction

Content

Use of indicative measurements and/or modelling to reduce fixed measurement network.

Remarks

Y/N.

FixedMeasurementReduction relates to the reduction of the number of sampling points for fixed measurements, which may be reduced by up to 50% under certain conditions as foreseen in the recasted Air Quality Directive 2024/2881 art. 9 §3.

In Reference

View reference attribute

ARZ_16 – ZoneResidentPopulationYear

Content

Reference year for the resident population data in the zone.

In Reference

View reference attribute

ARZ_17 – ZoneResidentPopulation

Content

Number of people residing within the air quality zone.

In Reference

View reference attribute

ARZ_18 – ClassificationYear

Content

The year of the last classification of the pollution level in the zone.

In Reference

View reference attribute

ARZ_19 – ClassificationDocumentId

Content

Identifier of the report detailing the classification procedure given by data provider.

In Reference

View reference attribute

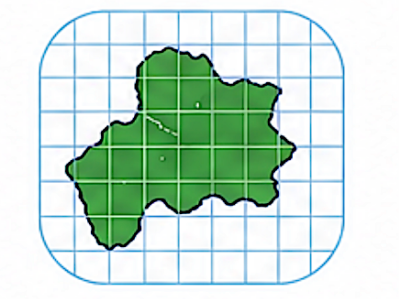
Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
ARZ_01	CountryCode	DU	varchar(2)	string	✓	✓
ARZ_02	AssessmentRegimeId	ARE_ZON_DU000A_0005_LV_H_aMean_2021_1	varchar(50)	string	✓	
ARZ_03	ZoneId	ZON_DU000A	varchar(50)	string		
ARZ_04	ZoneNationalCode	DU000A	varchar(50)	string		
ARZ_05	ZoneArea	442.31	decimal(10,2)	numeric		
ARZ_06	ZoneCategory	zone	varchar(20)	string		✓
ARZ_07	ZoneType	agg	varchar(20)	string		✓
ARZ_08	ZoneName	EcoDome Heights	varchar(150)	string		
ARZ_09	PollutantId	5	int	numeric		✓
ARZ_10	ProtectionTarget	H	varchar(50)	string		✓
ARZ_11	ObjectiveType	LV	varchar(50)	string		✓
ARZ_12	ReportingMetric	aMean	varchar(50)	string		✓
ARZ_13	AssessmentThresholdExceedance	aboveUAT	varchar(20)	string		✓
ARZ_14	PostponementYear	null	int	numeric		
ARZ_15	FixedMeasurementReduction	N	bit	boolean		
ARZ_16	ZoneResidentPopulationYear	2020	int	numeric		
ARZ_17	ZoneResidentPopulation	599159	int	numeric		
ARZ_18	ClassificationYear	2021	int	numeric		
ARZ_19	ClassificationDocumentId	DOC_ARE_ZON_DU_2021	varchar(150)	string		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

ZoneGeometry



Description

EEA extensions

ZoneGeometryGrid

Data management rules

Attributes

Attribute details

ZGE_01 – CountryCode

ZGE_02 – ZoneId

ZGE_03 – ZoneGeometryGeoJson

Example

Description

The purpose of the ZoneGeometry table is to collect information on air quality geometries.

ZoneGeometry table has to be reported whenever new information or an update of existing information is available. Attributes which create unique identifier of each record: **CountryCode** and **ZoneId**.

Updates of the ZoneGeometry table will not be time stamped - i.e. each update generates the only version of the table in reference/legacy data:

either as addition of a new zone geometry;

or modification of an existing geometry.

Validity of a new record will be tested using QC procedures (rules to be established), for example checking OGC compliance of the reported geometry.

Validity of modifications of existing geometries will also be tested through QC procedures. Allowed modifications are not supposed to affect outcomes of compliance analysis and/or assessments and will therefore be strictly limited to fulfil certain conditions (e.g. area difference thresholds).

Data products for the public will present records validated at the latest deadline. Data products for reporters will present records as most recently reported, highlighting differences between validated reports and recent reports in case of re-submissions.

EEA extensions

ZoneGeometry table will be extended by EEA in the reference/legacy data flow with a **Country** attribute.

In addition, a gridded version of the zones will be generated:

ZoneGeometryGrid

However, another gridded version of the zones will be generated in there - ZoneGeometryGrid, which will contain gridded AQ values in 2 layers:

1 Very High Resolution layer (100m) for populated areas,

1 High Resolution layer (1000m) for all areas.

ZoneGeometryGrid will be extended by EEA with the following attributes:

CountryCode;

ZoneId;

X, Y coordinates (SRID 3035);

Grid number identifiers based on X and Y coordinates.

Data management rules

It is recommended to report the ZoneGeometry table only for new zone geometries.

Zone geometries should be reported as geojson with SRID identifier and only the spatial data type “multipolygon” can be used. Expected projections are:

SRID3035,

SRID4258,

SRID4326 (the only allowed for the French overseas areas).

Records in the ZoneGeometry table (and subsequently in ZoneGeometryGrid) which do not have any correspondence to records in the AssessmentRegimeZone table will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
ZGE_01	CountryCode	varchar(2)	string	PK	countries	
ZGE_02	ZoneId	varchar(20)	string	PK		AssessmentRegimeZone
ZGE_03	ZoneGeometryGeoJson	varbinary	geometry			

Attribute details

ZGE_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

ZGE_02 – ZoneId

Content

Identifier of the air quality zone, given by data provider.

In Reference

View reference attribute

ZGE_03 – ZoneGeometryGeoJson

Content

Geospatial representation of the air quality zone (e.g. polygon geometry).

Remarks

It is allowed to report zone geometries in coordinate systems with the following EPSG codes:

3035

4258

4326

In Reference

View reference attribute

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
ZGE_01	CountryCode	DU	varchar(2)	string	✓	✓
ZGE_02	ZoneId	ZON_DU000A	varchar(20)	string	✓	
ZGE_03	ZoneGeometryGeoJson	{“type”:”Feature”,”geometry”: {“type”:”MultiPolygon”,”coordinates”: [[[[4.419074,51.3040095], [4.4181004999999995,51.305457],… ,[4.3362534,51.1730749], [4.3367638,51.1732234]]]},”properties”:{“srid”:”4326”}}	varbinary	geometry		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

ComplianceAssessmentMethod



Description

EEA extensions

Data management rules

Attributes

Attribute details

CAM_01 – CountryCode

CAM_02 – ReportingYear

CAM_03 – AssessmentRegimeId

CAM_04 – DataAggregationProcessId

CAM_05 – AssessmentMethodId

CAM_06 – PollutantId

CAM_07 – AssessmentType

CAM_08 – IsExceedance

CAM_09 – DataCoverage

CAM_10 – PollutionLevel

CAM_11 – PollutionLevelAdjusted

CAM_12 – RelativeUncertaintyLimit

CAM_13 – AssessmentMQI

CAM_14 – CorrectionFlag

CAM_15 – AttainmentId

CAM_16 – SRSId

CAM_17 – PreliminaryReason

CAM_18 – Deletion

Example

Description

The purpose of the ComplianceAssessmentMethod table is to collect information about assessment regimes and compliance situations on assessment method level. It specifies measured and/or modelled air quality values, aggregated in the context of the assessment regimes and allowed limits, provides uncertainty estimations for these values, describes some additional features of the air quality values (such as ‘correction flag’), links to spatial representativeness areas (for sampling points and possibly - exceedance values) and gives preliminary description of reason for exceedance (based on code listed values) if such occurs.

ComplianceAssessmentMethod table has to be reported every year. Attributes which create unique identifier of each record: CountryCode, AssessmentMethodId, DataAggregationProcessId and AttainmentId.

Updates of the ComplianceAssessmentMethod table will not be time stamped, i.e. each update generates the only version of the compliance situation that is stored at the EEA:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against aggregation code list, existing records in ModelObjectiveEstimation table, SamplingPoint table, SpatialRepresentativeness table and AssessmentRegimeZone table, as well as - indirectly - against ObservationMeasurementResult and MOEResult tables.

Valid modifications of existing record: beside the attributes which create unique identifier, as well as ReportingYear, AssessmentRegimeId, PollutantId and SRSId, modifications of value are in principle allowed for all other attributes. However, the degree of freedom for such modifications will be different, depending on data type and impact of modification, e.g.:

CorrectionFlag - all changes may be allowed,

Uncertainties - changes within certain numeric ranges,

PollutionLevel(s) - must agree with EEA's aggregations,

PreliminaryReason - all changes may be allowed within AttainmentId, but split into more compliance situation (> 1 of AttainmentId) will require additional rules (impact on PollutionLevelAdjustment, CompliancePlanLink), etc.

Validity of the modifications of existing record will be tested using QC (rules to be established). Allowed modifications may affect outcomes of compliance analysis and/or assessments, therefore re-submissions for past years, for which compliance reporting has been already done, will require additional tests vs reference/legacy data before official release of such data is granted.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported recently, highlighting the differences between validated reports and recent reports in case of re-submissions.

EEA extensions

ComplianceAssessmentMethod table will be extended by EEA by several attributes:

Country (for convenience), based on ISO2 country codes,

Pollutant (for convenience), based on code list,

SamplingPointReferenceId,

DataAggregationProcess, based on DataAggregationProcessId,

EEAAdjustmentEstimation, based on EEA's processing of information included in the PollutionLevelAdjustment table,

EEAExceedanceAssessment, based on EEAAdjustmentEstimation and allowed limit values,

AbsoluteUncertaintyLimit,

EEAMQIEstimation, based on EEA's interpretation of MOEResults vs. ObservationMeasurementResult.

Data management rules

In case of reporting AEI, one record for each relevant AssessmentRegimeId should be reported, specifying the value of AEI under PollutionLevel attribute and an OBE method identifier under AssessmentMethodId. OBE method (averaging across territorial unit) should be described in ModelObjectiveEstimation table with ResultEncoding 'zone'.

The values related to the new environmental objectives should be reported from the beginning of the new reporting in ReportNet3. Exceedances of the new thresholds should be indicated via IsExceedance attribute in the same way as exceedances of the currently used threshold values. They cannot and will not be handled as legally binding exceedances, i.e. will not determine compliance status of zones but they will be used for identification of needs for the roadmaps.

In preliminary reporting attributes such as CountryCode, AttainmentId, AssessmentMethodId, DataAggregationProcessId (PKs), as well as ReportingYear, AssessmentRegimeId and PollutantId must be populated, while other attributes may be provided with null values. Single AttainmentId for each ReportingYear and AssessmentRegimeId should be constructed, following rules given in the Identifiers section.

Deletion of an existing record is possible via the Deletion attribute (by reporting value = 1).

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
CAM_01	CountryCode	varchar(2)	string	PK	countries	
CAM_02	ReportingYear	int	numeric			
CAM_03	AssessmentRegimeId	varchar(100)	string			AssessmentRegimeZone
CAM_04	DataAggregationProcessId	varchar(50)	string	PK	aggregationprocess	
CAM_05	AssessmentMethodId	varchar(100)	string	PK		SamplingPoint SamplingProcess SamplingPointLocation ObservationMeasurementResult ModelObjectiveEstimation MOEResultInline MOEResultExternal
CAM_06	PollutantId	int	numeric		pollutant	
CAM_07	AssessmentType	varchar(50)	string		assessmenttype	
CAM_08	IsExceedance	varchar(5)	string			

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
CAM_09	DataCoverage	decimal(5,2)	numeric			
CAM_10	PollutionLevel	decimal(10,3)	numeric			
CAM_11	PollutionLevelAdjusted	decimal(10,3)	numeric			
CAM_12	RelativeUncertaintyLimit	decimal(10,2)	numeric			
CAM_13	AssessmentMQI	decimal(5,2)	numeric			
CAM_14	CorrectionFlag	bit	boolean			
CAM_15	AttainmentId	varchar(50)	string	PK		CompliancePlanLink PollutionLevelAdjustment
CAM_16	SRSId	varchar(50)	string			SpatialRepresentativeness
CAM_17	PreliminaryReason	varchar(50)	string		exceedancereason	
CAM_18	Deletion	bit	boolean			

Attribute details

CAM_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

CAM_02 – ReportingYear

Content

Year for which the data has been reported.

In Reference

View reference attribute

CAM_03 – AssessmentRegimeId

Content

Identifier of the air quality assessment regime, given by data provider.

Remarks

It will be cross-checked against the AssessmentRegimeZone table.

In Reference

View reference attribute

CAM_04 – DataAggregationProcessId

Content

Identifier of the process used for aggregating air quality data into statistical values.

In Reference

View reference attribute

CAM_05 – AssessmentMethodId

Content

Identifier of the assessment method used for air quality evaluation, given by data provider.

Remarks

Either SamplingPoint or Model/OBE, it will be cross-checked against the SamplingPoint table or the ModelObjectiveEstimation table, also - indirectly - against the ObservationMeasurementResult table and/or the ModellingResult tables.

In Reference

View reference attribute

CAM_06 – PollutantId

Content

Code of the air pollutant for which the assessment is being conducted.

In Reference

View reference attribute

CAM_07 – AssessmentType

Content

Classification of assessment methods into common types.

In Reference

View reference attribute

CAM_08 – IsExceedance

Content

Statement indicating whether pollution levels exceed environmental objectives.

In Reference

View reference attribute

CAM_09 – DataCoverage

Content

The proportion of the calendar year for which valid measurement data are available, expressed as a percentage.

Remarks

One element of DQO (see Directive 2024/2881 Section B.

In Reference

View reference attribute

CAM_10 – PollutionLevel

Content

Measured or modeled concentration level of the air pollutant.

Remarks

AirPollutionLevel must be reported for every AssessmentMethodId.

In Reference

View reference attribute

CAM_11 – PollutionLevelAdjusted

Content

Adjusted concentration level of the air pollutant, accounting for specific corrections.

In Reference

View reference attribute

CAM_12 – RelativeUncertaintyLimit

Content

The maximum relative uncertaintyfor assessment method (given for measurement).

Remarks

RelativeUncertaintyLimit must be reported for every AssessmentMethodId which refer to sampling points.

In Reference

View reference attribute

CAM_13 – AssessmentMQI

Content

Modelling Quality Indicator

In Reference

View reference attribute

CAM_14 – CorrectionFlag

Content

Correction factor for measured values applied (Y/N)

Remarks

Y/N. Correction is a string/boolean to be used in order to certify that, in case the SamplingProcess is equivalent to the reference method, the values have been corrected adequately (e.g. PM but also ozone with the cross section).

In Reference

View reference attribute

CAM_15 – AttainmentId

Content

Identifier of the air quality compliance situation, given by data provider (preliminary reason level).

Remarks

AttainmentId distinguishes each compliance situation. If there is no exceedance in the zone/assessment regime, there will be only one AttainmentId. If exceedance occurs, if it is caused (most probably) by the same reason and its extent (delineated by sampling point SRAs or modelling results) covers the whole zone, there will be also only one AttainmentId. However, if single exceedance (caused by the same reason) does not cover the whole zone, there will be 2 AttainmentIds - one for the exceedance situation and one for the non-exceedance situation in the same zone/assessment regime. Also - if the exceedances in the same zone/assessment regime are caused by different reasons, there will be more than 1 AttainmentIds.

In Reference

View reference attribute

CAM_16 – SRSId

Content

Identifier of the area representing spatial representativeness.

Remarks

Identifier linking to either SPO representativeness or to exceedance extent.

In Reference

View reference attribute

CAM_17 – PreliminaryReason

Content

Initial justification or reasoning for reported exceedance levels.

Remarks

This is crucial for distinguishing between different compliance situations in the same zone/assessment regime and pointing the relevant actions (air quality plans). It does not have to be precise at the first assessment of new exceedance and it can be changed from one year to another.

In Reference

View reference attribute

CAM_18 – Deletion

Content

Flag to indicate that this element and all related information must be deleted.

Remarks

Y/N

In Reference

View reference attribute

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
CAM_01	CountryCode	DU	varchar(2)	string	✓	✓

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
CAM_02	ReportingYear	2024	int	numeric		
CAM_03	AssessmentRegimeId	ARE_ZON_DU000A_0005_LV_H_aMean_2021_1	varchar(50)	string		
CAM_04	DataAggregationProcessId	P1Y	varchar(50)	string	✓	✓
CAM_05	AssessmentMethodId	SPO_DU0001_0005_100	varchar(100)	string	✓	
CAM_06	PollutantId	5	int	numeric		✓
CAM_07	AssessmentType	fixed	varchar(50)	string		✓
CAM_08	IsExceedance	FALSE	varchar(5)	string		
CAM_09	DataCoverage	85.78	decimal(5,2)	numeric		
CAM_10	PollutionLevel	21	decimal(10,3)	numeric		
CAM_11	PollutionLevelAdjusted	null	decimal(10,3)	numeric		
CAM_12	RelativeUncertaintyLimit	0.2	decimal(10,2)	numeric		
CAM_13	AssessmentMQI	null	decimal(5,2)	numeric		
CAM_14	CorrectionFlag	Y	bit	boolean		
CAM_15	AttainmentId	ATT_ZON_DU000A_00005_LV_H_aMean_2024_1	varchar(50)	string	✓	
CAM_16	SRSId	SRS_ZON_DU000A_00005_1	varchar(50)	string		
CAM_17	PreliminaryReason	null	varchar(50)	string		✓
CAM_18	Deletion	0	bit	boolean		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

SpatialRepresentativeness



Description

EEA extensions

Data management rules

Attributes

Attribute details

SRS_01 – CountryCode

SRS_02 – SRSId

SRS_03 – SRSApplicationId

SRS_04 – SRSApplication

SRS_05 – ResultEncoding

SRS_06 – RepresentativenessAssessmentMethodId

Example

Description

The purpose of the SpatialRepresentativeness table is to ensure the link between the coverage of the spatial representativeness area of a sampling point or exceedance extent, and the ComplianceAssessmentMethod table. It can also point to the method (model/OBE) used for the assessment of the SPO representativeness or the exceedance extent.

SpatialRepresentativeness table has to be reported whenever a new information or an update of existing information is available (at least every 5 years for the SPO representativeness and for each sampling point declared for compliance assessments). Attributes which create unique identifier of each record: CountryCode, SRSId and SRSApplicationId.

Updates of the SpatialRepresentativeness table will not be time stamped, i.e. each update generates the only version of the SpatialRepresentativeness that is stored at the EEA. Validity of a new record or modification will be tested using QC (rules to be established) - e.g. cross checking against existing records in ComplianceAssessmentMethod table, SRSInline and/or SRSExternal tables.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported the most recently.

EEA extensions

SpatialRepresentativeness table will be extended by EEA by:

Country (for convenience), based on ISO2 country codes.

Data management rules

It is allowed to reuse SRSId within the same AQ zone and across the years. However, it is strongly recommended to set a new SRSId in ComplianceAssessmentMethod table and new record in SpatialRepresentativeness table with SRSApplication = `exc_sr`, whenever a new exceedance is reported (also by model/OBE). This practice enables reporting of exceedance extents after relevant plans are in place, without re-submitting records for ComplianceAssessmentMethod table (but simply by using the SRSId which is already there). RepresentativenessAssessmentMethodId can be used for pointing to AssessmentMethodId in the ModelObjectiveEstimation table for definition of the modelling/OBE method which generated the SR areas (either for sampling point or as exceedance extents). It is expected that in case of exceedance extents, the value in RepresentativenessAssessmentMethodId will be the same as value in ScenarioAssessmentMethodId in PlanScenario record referring to the same AttainmentId (because it is the same model/OBE).

Records in SpatialRepresentativeness table which do not have any correspondence to records in ComplianceAssessmentMethods table will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SRS_01	CountryCode	varchar(2)	string	PK	countries	
SRS_02	SRSId	varchar(50)	string	PK		ComplianceAssessmentMethod
SRS_03	SRSApplicationId	varchar(50)	string	PK		SRSInline SRSExternal
SRS_04	SRSApplication	varchar(50)	string		SRapplication	
SRS_05	ResultEncoding	varchar(10)	string		resultencoding	

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SRS_06	RepresentativenessAssessmentMethodId	varchar(100)	string			ModelObjectiveEstimation

Attribute details

SRS_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

SRS_02 – SRSId

Content

Identifier of the area representing the sampling point’s measurement spatial coverage or the extent of the exceedance, given by data provider.

In Reference

View reference attribute

SRS_03 – SRSApplicationId

Content

Identifier pointing to the specific spatial representativeness given by data provider.

Remarks

The same SRSId can have several SRSApplicationId (e.g. one for the SPO representativeness area and one for the exceedance extent).

In Reference

View reference attribute

SRS_04 – SRSApplication

Content

Application of spatial representativeness (SPO representativeness or the exceedance extent).

Remarks

New code list (SPO representativeness area, exceedance extent area).

In Reference

View reference attribute

SRS_05 – ResultEncoding

Content

Encoding method used for model results (`internal` or `external`).

In Reference

View reference attribute

SRS_06 – RepresentativenessAssessmentMethodId

Content

Identifier of the assessment method (model) used for defining the area (either representativeness or extent), given by data provider.

Remarks

AssessmentMethodId is a model or OBE identifier that points to the method (model/OBE in the ModelObjectiveEstimation table) which was used for the assessments of the SPO representativeness area or the exceedance extent.

In Reference

View reference attribute

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SRS_01	CountryCode	DU	varchar(2)	string	✓	✓
SRS_02	SRSId	SRS_ZON_DU000A_00005_1	varchar(50)	string	✓	
SRS_03	SRSApplicationId	SRS_SPO_DU0001_0005_100	varchar(50)	string	✓	
SRS_04	SRSApplication	spo_sr	varchar(50)	string		✓
SRS_05	ResultEncoding	inline	varchar(10)	string		✓
SRS_06	RepresentativenessAssessmentMethodId	MOD_DU_PM10_NAT_SR	varchar(100)	string		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

SRSInline



Description

EEA extensions

Data management rules

Attributes

Attribute details

SRI_01 – CountryCode

SRI_02 – SRSApplicationId

SRI_03 – X

SRI_04 – Y

SRI_05 – SpatialResolution

Example

Description

The purpose of the SRSInline table is to collect the representativeness area (either SPO representativeness or exceedance extent) as set of grid cells following the common grid approach (see Introduction), for each sampling point declared for compliance assessments.

Attributes which create unique identifier of each record: CountryCode, SRSApplicationId, X and Y.

Updates of the SRSInline table will not be time stamped - i.e. each update generates the only version of the table in reference/legacy data. Validity of the modifications of existing record and or new record will be tested using QC (rules to be established).

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported the most recently.

EEA extensions

Reference data flow will have two tables representing reporting of SR areas:

SRSInventory;

SRSGrid.

SRSInventory will collect information about count of values reported for each SR area and will contain a reference to raw data file. SRSGrid table will differ from the reporting data flow however it will contain gridded AQ values in 3 layers:

2 Very High Resolution layers (10 and 100m - depending on original resolutions) for populated areas and traffic links;

1 High Resolution layer (1000m) for all areas.

SRSGrid table will be extended by several attributes:

Country (for convenience), based on ISO2 country codes;

Grid number identifiers, based on X,Y (SRID 3035).

Data management rules

It is expected that SR areas for sampling points declared for compliance assessments will be reported together with attainment reports (when ComplianceAssessmentMethod table is reported). Exceedance extents are expected to be reported as soon as relevant air quality plans are in place.

Records in SRSInline table which do not have any correspondence to records in SpatialRepresentativeness table will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SRI_01	CountryCode	varchar(2)	string	PK	countries	
SRI_02	SRSApplicationId	varchar(50)	string	PK		SpatialRepresentativeness

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SRI_03	X	bigint	numeric	PK		
SRI_04	Y	bigint	numeric	PK		
SRI_05	SpatialResolution	int	numeric		spatialresolution	

Attribute details

SRI_01 – CountryCode

Content

Country or territory ISO2 code.

SRI_02 – SRSApplicationId

Content

Identifier of the specific spatial representativeness given by data provider.

SRI_03 – X

Content

X-coordinate of grid cell belonging to SRA in a projected coordinate reference system.

Remarks

It must be using SRID 3035.

SRI_04 – Y

Content

Y-coordinate of grid cell belonging to SRA in a projected coordinate reference system.

Remarks

It must be using SRID 3035.

SRI_05 – SpatialResolution

Content

Description of the spatial resolution of the data, indicating the level of detail.

Remarks

SpatialResolution will be tested against the code list for allowed values.

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SRI_01	CountryCode	DU	varchar(2)	string	✓	✓
SRI_02	SRSApplicationId	SRS_SPO_DU0001_0005_100	varchar(50)	string	✓	
SRI_03	X	3825000	bigint	numeric	✓	
SRI_04	Y	2274000	bigint	numeric	✓	
SRI_05	SpatialResolution	1000	int	numeric		✓

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

SRSExternal



Description

EEA extensions

Data management rules

Attributes

Attribute details

SRE_01 – CountryCode

SRE_02 – SRSApplicationId

SRE_03 – SpatialResolution

SRE_04 – GISAttachment

Example

Description

The purpose of the SRSExternal table is to collect the representativeness area (either SPO representativeness or exceedance extent) as external GIS files, for each sampling point declared for compliance assessments.

Attributes which create unique identifier of each record: CountryCode and SRSApplicationId.

Updates of the SRSExternal table will not be time stamped - i.e. each update generates the only version of the table in reference/legacy data. Validity of the modifications of existing record and/or new record will be tested using QC (rules to be established).

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported the most recently.

EEA extensions

Reference data flow will have two tables representing reporting of SR areas:

SRSInventory;

SRSGrid.

SRSInventory will collect information about count of values reported for each SR area and will contain a reference to raw data file. SRSGrid table will differ from the reporting data flow however it will contain gridded AQ values in 3 layers:

2 Very High Resolution layers (10 and 100m - depending on original resolutions) for populated areas and traffic links;

1 High Resolution layer (1000m) for all areas.

SRSGrid table will be extended by several attributes:

Country (for convenience), based on ISO2 country codes;

Grid number identifiers, based on X,Y (SRID 3035).

Data management rules

It is expected that SR areas for sampling points declared for compliance assessments will be reported together with attainment reports (when ComplianceAssessmentMethod table is reported).

Exceedance extents are expected to be reported as soon as relevant air quality plans are in place.

Records in SRSExternal table which do not have any correspondence to records in SpatialRepresentativeness table will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SRE_01	CountryCode	varchar(2)	string	PK	countries	
SRE_02	SRSApplicationId	varchar(50)	string	PK		SpatialRepresentativeness
SRE_03	SpatialResolution	int	numeric		spatialresolution	
SRE_04	GeoTiffAttachment	varchar(100)	attachment (R3 data type)			

Attribute details

SRE_01 – CountryCode

Content

Country or territory ISO2 code.

SRE_02 – SRSApplicationId

Content

Identifier of the specific spatial representativeness given by data provider.

SRE_03 – SpatialResolution

Content

Description of the spatial resolution of the data, indicating the level of detail.

Remarks

SpatialResolution will be tested against the code list for allowed values.

SRE_04 – GISAttachment

Content

Attached GIS file containing the spatial representativeness area.

Remarks

The attachment contains either:

sampling point representativeness area; or

exceedance extent area.

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SRE_01	CountryCode	DU	varchar(2)	string	✓	✓

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SRE_02	SRSApplicationId	SRS_SPO_DU0001_0008_100	varchar(50)	string	✓	
SRE_03	SpatialResolution	100	int	numeric		✓
SRE_04	GeoTiffAttachment		varchar(100)	attachment (R3 data type)		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

PollutionLevelAdjustment



Description

EEA extensions

Data management rules

Attributes

Attribute details

ADJ_01 – CountryCode

ADJ_02 – AttainmentId

ADJ_03 – AdjustmentSource

ADJ_04 – AdjustmentAssessmentMethodId

ADJ_05 – AdjustmentDocumentId

Example

Description

The purpose of the PollutionLevelAdjustment table is to collect information on deductions (adjustments) due to natural sources or winter salting and sanding. It describes the adjustment source/type (using code listed values) as well as points to the method (model/OBE) used for the assessments.

PollutionLevelAdjustment (record) table has to be reported whenever a new information or an update of existing information is available. Attributes which create unique identifier of each record: CountryCode, AttainmentId and AdjustmentSource.

Updates of the PollutionLevelAdjustment table will not be time stamped - i.e. each update generates the only version of the table in reference/legacy data:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against existing records in ModelObjectiveEstimation table and ComplianceAssessmentMethod table, as well as - indirectly - against MOEResult tables.

Valid modifications of existing record in this table are: minor corrections e.g. of wrong code list values. Validity of the modifications of existing record will be tested using QC (rules to be established). Allowed modifications are not supposed to directly affect outcomes of compliance analysis and/or assessments.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported recently, highlighting the differences between validated reports and recent reports in case of re-submissions.

EEA extensions

PollutionLevelAdjustment table will be extended by:

Country (for convenience), based on ISO2 country codes.

Data management rules

AdjustmentAssessmentMethodId must be used for pointing to AssessmentMethodId in the ModelObjectiveEstimation table for definition of the modelling/OBE method which generated the adjustment values.

AdjustmentAssessmentMethodId must be different for each AdjustmentSource so that it is possible to report adjustment values for each of them in the MOEResult tables.

Records in PollutionLevelAdjustment table which do not have any correspondence to records in ComplianceAssessmentMethods table will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
ADJ_01	CountryCode	varchar(2)	string	PK	countries	
ADJ_02	AttainmentId	varchar(100)	string	PK		ComplianceAssessmentMethod

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
ADJ_03	AdjustmentSource	varchar(50)	string	PK	adjustmentsource type	
ADJ_04	AdjustmentAssessmentMethodId	varchar(100)	string			ModelObjectiveEstimation
ADJ_05	AdjustmentDocumentId					Documentation

Attribute details

ADJ_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

ADJ_02 – AttainmentId

Content

Identifier of the air quality compliance situation, given by data provider.

Remarks

It will be cross-checked against the ComplianceAssessmentMethod table. This attribute must follow a defined structure (see Identifiers section).

In Reference

View reference attribute

ADJ_03 – AdjustmentSource

Content

Description of the source being adjusted (e.g., sea spray, volcanic activity).

In Reference

View reference attribute

ADJ_04 – AdjustmentAssessmentMethodId

Content

Identifier of the method - model - used for deduction assessment, given by data provider.

Remarks

AdjAssessmentMethodId is a model or OBE identifier that points to the method (model/OBE in the ModelObjectiveEstimation table) which was used for the assessments of the deduction. If there are several different adjustment types/sources, different AdjAssessmentMethodIds must be used, so that it is possible to distinguish between corresponding adjustment values reported in the ModellingResult (Inline or External) table. It will be cross-checked against the Model table, also - indirectly - against the ModellingResult tables.

In Reference

View reference attribute

ADJ_05 – AdjustmentDocumentId

In Reference

View reference attribute

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
ADJ_01	CountryCode	DU	varchar(2)	string	✓	✓
ADJ_02	AttainmentId	ATT_ZON_DU000B_00005_LV_H_aMean_2024_1	varchar(50)	string	✓	
ADJ_03	AdjustmentSource	H	varchar(50)	string	✓	✓
ADJ_04	AdjustmentAssessmentMethodId	OBE_DU_SEASALT_PM10_H_LV_ADJ	varchar(100)	string		
ADJ_05	AdjustmentDocumentId	null				

Legend: PK = Primary Key; FK = Foreign Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

CompliancePlanLink



Description

EEA extensions

Data management rules

Attributes

Attribute details

CPL_01 – CountryCode

CPL_02 – AttainmentId

CPL_03 – PlanId

CPL_04 – ScenarioId

CPL_05 – SourceApportionmentId

CPL_06 – PlanCategory

CPL_07 – PlanTitle

CPL_08 – PlanAdoptionDate

CPL_09 – PlanBeginDate

CPL_10 – PlanEndDate

CPL_11 – PlanDocumentId

CPL_12 – Deletion

Example

Description

The purpose of the CompliancePlanLink table is to ensure the link between the AttainmentId as reported in the ComplianceAssessmentMethod table with the plan developed to tackle the exceedance (with source apportionment and relevant scenario).

CompliancePlanLink table has to be reported every year to maintain the link between exceedance situations (evolving through years) and the relevant plans. Attributes which create unique identifier of each record: CountryCode, AttainmentId, PlanId, ScenarioId and SourceApportionmentId.

Updates of the CompliancePlanLink table will not be time stamped, i.e. each update generates the only version of the CompliancePlanLink that is stored at the EEA. Since all the attributes create unique identifier of each record, the only update possible for this table is addition of a new record. Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against existing records in ComplianceAssessmentMethod table, PlanScenario table and SourceApportionment table.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported recently, highlighting the differences between validated reports and recent reports in case of re-submissions.

EEA extensions

In the reference data flow, the CompliancePlanLink table will collect the attributes CountryCode, AttainmentId, PlanId, ScenarioId, SourceApportionmentID as well as Deletion. It will be extended be extended by:

Country (for convenience), based on ISO2 country codes,

ReportingYear (for convenience), based on ComplianceAssessmentMethod (year of AttainmentId).

The attributes CountryCode,PlanCategory, PlanTitle, PlanAdoptionDate, PlanBeginDate, PlanEndDate, PlanDocumentID and Deletion will be collected in the separate table Plan (not existing in the Reporting flow) in which Country will be added for convenience.

Data management rules

All AttainmentIds which refer to exceedance indicated in ComplianceAssessmentMethod must be covered by plan developed to tackle the exceedance (with source apportionment and relevant scenario). There can be only one unique combination of PlanId, ScenarioId and SourceApportionmentId for each AttainmentId. However, there can be many AttainmentIds (describing either one or many compliance situations evolving across years) for each unique combination of PlanId, ScenarioId and SourceApportionmentId.

It is recommended not to change PlanDocumentId in case of documentation update related to the same plan/scenario. Instead, a new attachment should be uploaded in Document table re-using the same DocumentId. Previous versions of documentation will be kept in reference data. In case when the same PlanDocumentId was used for all records and there is a need to modify documentation e.g. only for one specific scenario, it is recommended to create a new PlanDocumentId for that record in CompliancePlanLink table, and then report old (plan specific) and new (e.g. scenario specific) documents as attachments for that new PlandocumentId (as DocumentId) in the Document table in a subsequent manner.

Records in CompliancePlanLink table which do not have any correspondence to records in ComplianceAssessmentMethods table will be flagged for deletion in the reference data flow.

Deletion of an existing record will be also possible via the Deletion attribute (by reporting value = 1).

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
CPL_01	CountryCode	varchar(2)	string	PK	countries	
CPL_02	AttainmentId	varchar(50)	string	PK		ComplianceAssessmentMethod
CPL_03	PlanId	varchar(50)	string	PK		PlanScenario
CPL_04	ScenarioId	varchar(50)	string	PK		PlanScenario
CPL_05	SourceApportionmentId	varchar(50)	string	PK		SourceApportionment
CPL_06	PlanCategory	varchar(20)	string		plancategory	
CPL_07	PlanTitle	nvarchar(1000)	string			
CPL_08	PlanAdoptionDate	date	date			
CPL_09	PlanBeginDate	date	date			
CPL_10	PlanEndDate	date	date			
CPL_11	PlanDocumentId	varchar(150)	string			Documentation
CPL_12	Deletion	bit	boolean			

Note

Attributes with ReportNet3 data type `date` or `datetime` shall use the ISO 8601 format. Date-time values may include a local UTC offset.

Attribute details

CPL_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

CPL_02 – AttainmentId

Content

Identifier of the air quality compliance situation, given by data provider. This attribute must follow a defined structure (see Identifiers section).

Remarks

Several AttainmentIds can point to the same PlanId (e.g. the plan covers different pollutants or successive years of exceedance for the same pollutant). Alternatively, there might be several plans applying to the same AttainmentId (e.g. different authority level, or successive plans). It will be cross-checked against the ComplianceAssessmentMethod table.

In Reference

View reference attribute

CPL_03 – PlanId

Content

Identifier of the air quality plan, given by data provider. This attribute must follow a defined structure (see Identifiers section).

Remarks

It will be cross-checked against the PlanScenario table.

In Reference

View reference attribute

CPL_04 – ScenarioId

Content

Identifier of the scenario, given by data provider.

Remarks

ScenarioId, several scenarios can apply to the same PlanId (e.g. different pollutant and/or data aggregation) and vice versa. It will be cross-checked against the PlanScenario table. This attribute must follow a defined structure (see Identifiers section).

In Reference

View reference attribute

CPL_05 – SourceApportionmentId

Content

Identifier of the source apportionment, given by data provider.

Remarks

SourceApportionmentId, there might be only one SourceApportionment per ScenarioId but several per PlanId (e.g.: different pollutant and/or data aggregation). It will be cross-checked against the SourceApportionment table.

In Reference

View reference attribute

CPL_06 – PlanCategory

Content

Category of the air quality plan (e.g., roadmap, short-term plan, plan).

Remarks

PlanCategory: corresponds to the different type of plan considered under the recasted Air Quality Directive 2024/2881 – roadmap, short-term plan, plan.

CPL_07 – PlanTitle

Content

Title of the plan

Remarks

PlanTitle can be reported in any of the EU languages.

CPL_08 – PlanAdoptionDate

Content

Date of official adoption of the plan by the responsible authorities

CPL_09 – PlanBeginDate

Content

Start date (enforcement date) for the plan

Remarks

Begin/End date refers to the validity period of the plan/roadmap, i.e. when it enters into force and until it’s unvalid. Note that some plans might not have an end date. In such cases PlanEndDate should be NULL.

CPL_10 – PlanEndDate

Content

End date for the plan

Remarks

Begin/End date refers to the validity period of the plan/roadmap, i.e. when it enters into force and until it’s unvalid. Note that some plans might not have an end date. In such cases PlanEndDate should be NULL.

CPL_11 – PlanDocumentId

Content

Identifier of the air quality plan document given by data provider.

CPL_12 – Deletion

Content

Flag to indicate that this element must be deleted

Remarks

Y/N

In Reference

View reference attribute

Example

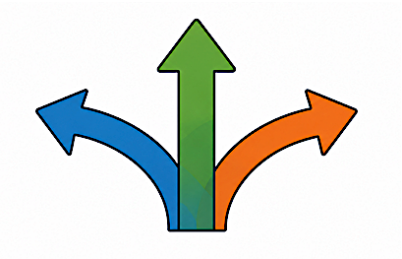
Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
CPL_01	CountryCode	DU	varchar(2)	string	✓	✓
CPL_02	AttainmentId	ATT_ZON_DU000B_00005_LV_H_daysAbove_2024_1	varchar(50)	string	✓	
CPL_03	PlanId	PLA_ZON_DU000B_00005_LV_H_daysAbove	varchar(50)	string	✓	
CPL_04	ScenarioId	SCE_ZON_DU000B_00005_LV_H_daysAbove_1	varchar(50)	string	✓	
CPL_05	SourceApportionmentId	SAP_ZON_DU000B_00005_LV_H_daysAbove	varchar(50)	string	✓	
CPL_06	PlanCategory	plan	varchar(20)	string		✓
CPL_07	PlanTitle	აირე კალიტატე პროგრამა გენერალა, კლიმა ბაბეს ე ემისა ულიე პლანა პორ ურბა-იუგ დუსტოვა რეჟიონ, მე ვეპრიმ პლანა, ტრეგუეს ე ნდიეკიე მეხანიზამ 2028–2035	nvarchar(1000)	string		
CPL_08	PlanAdoptionDate	2025-11-28	date	date		
CPL_09	PlanBeginDate	2026-01-01	date	date		
CPL_10	PlanEndDate	2030-12-31	date	date		

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
CPL_11	PlanDocumentId	DOC_ZON_DU000B_00005_LV_H_daysAbove	varchar(150)	string		
CPL_12	Deletion	0	bit	boolean		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

PlanScenario



Description

EEA extensions

Data management rules

Attributes

Attribute details

PSC_01 – CountryCode

PSC_02 – PlanId

PSC_03 – ScenarioId

PSC_04 – ScenarioCategory

PSC_05 – ScenarioNationalCode

PSC_06 – PollutantId

PSC_07 – DataAggregationProcessId

PSC_08 – ScenarioYear

PSC_09 – ScenarioPollutionLevel

PSC_10 – ExposedPopulation

PSC_11 – ScenarioAssessmentMethodId

Example

Description

The purpose of the PlanScenario table is to provide information on the plan and on the scenario associated to the plan. It is the way of reporting combined impact of measures (which could originate from different plans - e.g. on national, regional and local levels) on a specific compliance situation (which may consist of several AttainmentIds - describing evolution over time in one or many locations). It specifies plan category (plan, roadmap, short-term) and core information regarding scenarios such as: category (reference, baseline, mitigation), year, projected air pollution levels, and model/OBE which was used for generating the scenario. It also gives further details about the plan and scenarios through attached documentation.

PlanScenario (record) table has to be reported whenever a new information or an update of existing information is available. Attributes which create unique identifier of each record: CountryCode, PlanId, ScenarioId and ScenarioCategory.

Updates of the PlanScenario table will not be time stamped i.e. each update generates the only version of the PlanScenario that is stored at the EEA:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against relevant code lists, it may be cross-checking against the Model table.

Valid modifications of existing record in this table are:

almost all allowed (apart from ScenarioAssessmentMethodId which must agree with AssessmentMethodId in ModelObjectiveEstimation table) within the scope of the code lists and their potential interconnections.

Validity of the modifications of existing record will be tested using QC (rules to be established). Allowed modifications may affect outcomes of scenario analysis and/or assessments of measures, therefore re-submissions for past years, for which plans reporting has been already done, will require additional tests vs reference/legacy data before official release of such data is granted.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported recently, highlighting the differences between validated reports and recent reports in case of re-submissions.

EEA extensions

PlanScenario table will be extended by several attributes:

Country (for convenience), based on ISO2 country codes,

Pollutant (for convenience), based on code list,

DataAggregationProcess, based on DataAggregationProcessId,

EEAExposedPopulation, based on EEA's estimation of population exposed to exceedance in scenario year.

Data management rules

ScenarioAssessmentMethodId must be used for pointing to AssessmentMethodId in the ModelObjectiveEstimation table for definition of the modelling/OBE method which generated the scenario values. ScenarioAssessmentMethodId must be different for each ScenarioId and ScenarioCategory so that it is possible to report values for each of them in the MOEResult tables. One of the MOEResult tables (inline or external) must be populated for each ScenarioAssessmentMethodId but the results may be reported just for the grid cells corresponding to the sampling point locations.

It is expected that modelling will be used for delineation of exceedance extents, including the 'reference' scenario category. However, if exceedance was reported based on sampling point measurements, the values reported in MOEResult tables for the 'reference' scenario should equal the measured values in the grid cells corresponding to the sampling point locations. ScenarioPollutionLevel must agree with the maximum value reported in MOEResult tables under model/OBE declared as ScenarioAssessmentMethodId.

Records in PlanScenario table which do not have any correspondence to records in CompliancePlanLink table or ScenarioMeasure table will be flagged for deletion in the reference data flow.

Keep in mind that authorities responsible for establishing plans and for reporting the plans to the EU should be reported using Authority table.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
PSC_01	CountryCode	varchar(2)	string	PK	countries	
PSC_02	PlanId	varchar(50)	string	PK		CompliancePlanLink

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
PSC_03	ScenarioId	varchar(50)	string	PK		CompliancePlanLink ScenarioMeasure
PSC_04	ScenarioCategory	varchar(20)	string	PK	scenariocategory	
PSC_05	ScenarioNationalCode	varchar(50)	string			
PSC_06	PollutantId	int	numeric		pollutant	
PSC_07	DataAggregationProcessId	varchar(50)	string		aggregationprocess	
PSC_08	ScenarioYear	int	numeric			
PSC_09	ScenarioPollutionLevel	decimal(10,2)	numeric			
PSC_10	ExposedPopulation	int	numeric			
PSC_11	ScenarioAssessmentMethodId	varchar(100)	string			ModelObjectiveEstimation

Attribute details

PSC_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

PSC_02 – PlanId

Content

Identifier of the air quality plan, given by data provider. This attribute must follow a defined structure (see Identifiers section).

In Reference

View reference attribute

PSC_03 – ScenarioId

Content

Identifier of the scenario, given by data provider. This attribute must follow a defined structure (see Identifiers section).

In Reference

View reference attribute

PSC_04 – ScenarioCategory

Content

Classification of the scenario (e.g., baseline, projection).

Remarks

ScenarioCategory: reference, baseline or projection. It should be included in the composite PK in the next iteration of schema version.

In Reference

View reference attribute

PSC_05 – ScenarioNationalCode

Content

Unique local code assigned to the evaluation scenario by the data provider.

In Reference

View reference attribute

PSC_06 – PollutantId

Content

Code of the air pollutant being considered in the scenario.

Remarks

AirPollutantCode: must correspond to the AirPollutantCode of the AttainmentId.

In Reference

View reference attribute

PSC_07 – DataAggregationProcessId

Content

Identifier for the process used to aggregate air quality data in the scenario.

Remarks

DataAggregationProcessId: the data aggregation used for the scenario; it must correspond to the DataAggregationProcessId of the AttainmentId.

In Reference

View reference attribute

PSC_08 – ScenarioYear

Content

The calendar year for which the scenario has been modeled.

In Reference

View reference attribute

PSC_09 – ScenarioPollutionLevel

Content

Estimated air pollution level in the scenario year.

In Reference

View reference attribute

PSC_10 – ExposedPopulation

Content

PSC_11 – ScenarioAssessmentMethodId

In Reference

View reference attribute **Content**

Identifier of the assessment method - model - used in the scenario, given by data provider.

Remarks

AssessmentMethodId: the model/OBE used for producing the scenario (also to be declared in the ModelObjectiveEstimation table). It may be cross-checked against Model table, also - indirectly - against ModellingResult table.

In Reference

View reference attribute

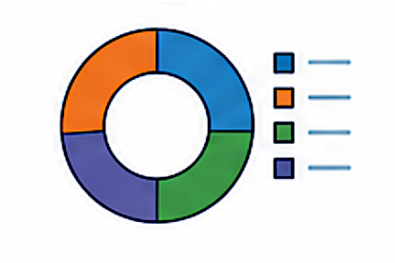
Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
PSC_01	CountryCode	DU	varchar(2)	string	✓	✓
PSC_02	PlanId	PLA_ZON_DU000B_00005_LV_H_daysAbove	varchar(50)	string	✓	
PSC_03	ScenarioId	SCE_ZON_DU000B_00005_LV_H_daysAbove	varchar(50)	string	✓	
PSC_04	ScenarioCategory	reference	varchar(20)	string	✓	✓
PSC_05	ScenarioNationalCode	PLA_another_scenario	varchar(50)	string		
PSC_06	PollutantId	5	int	numeric		✓
PSC_07	DataAggregationProcessId	P1Y-daysAbove50	varchar(50)	string		✓
PSC_08	ScenarioYear	2024	int	numeric		
PSC_09	ScenarioPollutionLevel	43	decimal(10,2)	numeric		
PSC_10	ExposedPopulation	95250	int	numeric		
PSC_11	ScenarioAssessmentMethodId	MOD_DU_PM10_LOC_SCEN_REF	varchar(100)	string		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

SourceApportionment



Description

EEA extensions

Data management rules

Attributes

Attribute details

SAP_01 – CountryCode

SAP_02 – SourceApportionmentId

SAP_03 – Contribution Type

SAP_04 – SpatialScale

SAP_05 – SourceSector

SAP_06 – PollutantId

SAP_07 – Contribution

SAP_08 – SourceApportionmentDocumentId

Example

Description

The purpose of the SourceApportionment table is to report information on the sources contributing to air pollution exceedances. It describes the spatial scale of the contributions, their type (background, increment) as well as sector of origin.

SourceApportionment (record) table has to be reported whenever a new source apportionment has been done (new AttainmentId) or an update of existing information is available (existing AttainmentId). Attributes which create unique identifier of each record: CountryId, SourceApportionmentId, ContributionType, SpatialScale and SourceSector.

Updates of the SourceApportionment table will not be time stamped, i.e. each update generates the only version of the SourceApportionment that is stored at the EEA:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against relevant code lists.

Valid modifications of existing record in this table are:

all allowed within the scope of the code lists and their potential interconnections.

Validity of the modifications of existing record will be tested using QC (rules to be established - e.g. Contribution must be =< 100%).

Allowed modifications may affect outcomes of scenario analysis and/or assessments of measures, therefore re-submissions for past years, for which plans reporting has been already done, will require additional tests vs reference/legacy data before official release of such data is granted.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported recently, highlighting the differences between validated reports and recent reports in case of re-submissions.

EEA extensions

SourceApportionment table will be extended by:

Country (for convenience), based on ISO2 country codes;

Pollutant (for convenience), based on code list.

Data management rules

It is recommended not to change SourceApportionmentDocumentId in case of documentation update related to the same plan/scenario. Instead, a new attachment should be uploaded in Document table re-using the same DocumentId. Previous versions of documentation will be kept in reference data. In case when the same SourceApportionmentDocumentId was used for all records and there is a need to modify documentation only for one specific source apportionment, it is recommended to create a new SourceApportionmentDocumentId for that record in SourceApportionment table, and then report old (general) and new (source apportionment specific) documents as attachments for that new SourceApportionmentDocumentId (as DocumentId) in the Document table in a subsequent manner.

Records in SourceApportionment table which do not have any correspondence to Plans and Scenarios in CompliancePlanLink table will be flagged for deletion in the reference data flow. However, it is also possible to set the records for deletion by reporting Contribution value as -1.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SAP_01	CountryCode	varchar(2)	string	PK	countries	
SAP_02	SourceApportionmentId	varchar(50)	string	PK		ComplianceAssessmentMethod
SAP_03	ContributionType	varchar(20)	string	PK	contributiontype	
SAP_04	SpatialScale	varchar(50)	string	PK	spatialscale	
SAP_05	SourceSector	varchar(50)	string	PK	sourcesectors	
SAP_06	PollutantId	int	numeric		pollutant	
SAP_07	Contribution	decimal(8,2)	numeric			
SAP_08	SourceApportionmentDocumentId	varchar(150)	string			Documentation

Attribute details

SAP_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

SAP_02 – SourceApportionmentId

Content

Identifier of the source apportionment, given by data provider.

In Reference

View reference attribute

SAP_03 – Contribution Type

Content

Type of contribution (e.g. background, increment).

In Reference

View reference attribute

SAP_04 – SpatialScale

Content

Geographical scope of the contribution (e.g., urban, local, regional, national).

In Reference

View reference attribute

SAP_05 – SourceSector

Content

The sector responsible for emissions (e.g., traffic, industry, residential heating).

In Reference

View reference attribute

SAP_06 – PollutantId

Content

Code of the air pollutant for which the contribution is being assessed.

Remarks

AirPollutantCode: must correspond to the AirPollutantCode of the AttainmentId.

In Reference

View reference attribute

SAP_07 – Contribution

Content

Estimated contribution of the specified source sector to air pollution levels [%].

Remarks

The value is understood as applicable to AirPollutionLevel adjusted for natural sources. Therefore contributions should not include the natural sources, should be given as % and sum up to 100% within SourceAppId.

In Reference

View reference attribute

SAP_08 – SourceApportionmentDocumentId

Content

Identifier of the documentation on source apportionment.

In Reference

View reference attribute

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SAP_01	CountryCode	DU	varchar(2)	string	✓	✓
SAP_02	SourceApportionmentId	SAP_ZON_DU000B_00005_LV_H_daysAbove	varchar(50)	string	✓	

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SAP_03	ContributionType	background	varchar(20)	string	✓	✓
SAP_04	SpatialScale	regional	varchar(50)	string	✓	✓
SAP_05	SourceSector	other	varchar(50)	string	✓	✓
SAP_06	PollutantId	5	int	numeric		✓
SAP_07	Contribution	9	decimal(8,2)	numeric		
SAP_08	SourceApportionmentDocumentId	DOC_SAP_ZON_DU000B_00005_LV_H_daysAbove	varchar(150)	string		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

ScenarioMeasure



Description

EEA extensions

Data management rules

Attributes

Attribute details

SME_01 – CountryCode

SME_02 – ScenarioId

SME_03 – ScenarioCategory

SME_04 – MeasureGroupId

SME_05 – MeasureGroupPollutionReduction

SME_06 – MeasureReductionAssessmentMethodId

Example

Description

The purpose of the ScenarioMeasure table is to define the group(s) of measures (MeasureGroupId) associated to a scenario, as well as the total (by group of measures) reduction of pollution level which can be expected. It also points to model/OBE which was used for generating the results for the group of measures.

ScenarioMeasure (record) table has to be reported whenever a new information or an update of existing information is available. Attributes which create unique identifier of each record: CountryCode, ScenarioId, MeasureGroupId and ScenarioCategory.

Updates of the ScenarioMeasure table will not be time stamped, i.e. each update generates the only version of the ScenarioMeasure that is stored at the EEA:

either as addition of new record, understood as addition of new combination of values for the attributes creating unique identifier,

or modification of existing record, understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against PlanScenario table and Measure table, it may be cross-checking against the ModelObjectiveEstimation table.

There are no valid modifications of existing record in this table.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported recently, highlighting the differences between validated reports and recent reports in case of re-submissions.

EEA extensions

ScenarioMeasure table will be extended by several attributes:

Country (for convenience), based on ISO2 country codes,

PollutantId (for convenience), from PlanScenario for the corresponding ScenarioId,

Pollutant (for convenience), based on code list,

DataAggregationProcessId, (for convenience), from PlanScenario for the corresponding ScenarioId,

DataAggregationProcess, (for convenience) based on DataAggregationProcessId.

Data management rules

MeasureGroupId allows reporting of impacts for group of measures e.g. originating from different national, regional or local level plans. In such case - MeasureReductionAssessmentMethodId may be used for pointing to AssessmentMethodId in the ModelObjectiveEstimation table for definition of the modelling/OBE method which generated the scenario values for each measure group. In case the MeasureReductionAssessmentMethodId is reported - it must be different for each ScenarioId, ScenarioCategory and MeasureGroupId so that it is possible to report values for each of them in the MOEResult tables. Also, one of the MOEResult tables must be populated for each reported MeasureReductionAssessmentMethodId but the results may be provided just for the grid cells corresponding to the sampling point locations.

There is no need to report records for ‘reference’ scenario category in ScenarioMeasure table.

If MeasureGroupPollutionReduction is reported - the sum for specific scenario category within ScenarioId must be =< the difference between the values of ScenarioPollutionLevel reported for ‘reference’ and that ScenarioId and ScenarioCategory in PlanScenario table. It is not allowed to re-use MeasureGroupId in different ScenarioId (to avoid complications in deletion procedures).

Records in ScenarioMeasure table which do not have any correspondence to records in PlanScenario table or Measure table will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
SME_01	CountryCode	varchar(2)	string	PK	countries	
SME_02	ScenarioId	varchar(50)	string	PK		ComplianceAssessmentMethod PlanScenario
SME_03	ScenarioCategory	varchar(20)	string	PK	scenariocategory	
SME_04	MeasureGroupId	varchar(50)	string	PK		MeasurementStation
SME_05	MeasureGroupPollutionReduction	decimal(10,2)	numeric			
SME_06	MeasureReductionAssessmentMethodId	varchar(100)	string			ModelObjectiveEstimation

Attribute details

SME_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

SME_02 – ScenarioId

Content

Identifier of the scenario, given by data provider. This attribute must follow a defined structure (see Identifiers section).

Remarks

It will be cross-checked against PlanScenario table.

In Reference

View reference attribute

SME_03 – ScenarioCategory

Content

Classification of the scenario (e.g., baseline, projection).

Remarks

ScenarioCategory: baseline or projection. It should be included in the composite PK in the next iteration of schema version.

In Reference

View reference attribute

SME_04 – MeasureGroupId

Content

Identifier of the pollution reduction for measure group, given by data provider.

Remarks

MeasureGroupId: distinct groups might contain one or more measures. For example, group 1 contains all the measures related to traffic, while group 2 contains all the measures related to industry. To the extreme, the group may correspond to one measure only; the same MeasureGroupId might be used for different scenarios, also - several MeasureGroupId might be declared for the same scenario.

It will be cross-checked against Measure table.

In Reference

View reference attribute

SME_05 – MeasureGroupPollutionReduction

Content

Reduction in air pollution concentration levels due to the applied group of measures.

Remarks

Sum of MeasureGroupAirPollutionReduction within ScenarioId should agree with the difference between value of AirPollutionLevel in the first AttainmentId and the value reported in PlanScenario as ScenarioAirPollutionLevel.

In Reference

View reference attribute

SME_06 – MeasureReductionAssessmentMethodId

Content

Identifier of the assessment method - model - used in the scenario, given by data provider.

Remarks

AssessmentMethodId: the model/OBE used for producing the results for the measure group (also to be declared in the ModelObjectiveEstimation table). It may be cross-checked against Model table, also - indirectly - against ModellingResult table.

In Reference

View reference attribute

Example

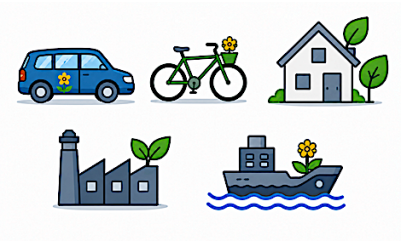
Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SME_01	CountryCode	DU	varchar(2)	string	✓	✓
SME_02	ScenarioId	SCE_ZON_DU000B_00005_LV_H_daysAbove	varchar(50)	string	✓	
SME_03	ScenarioCategory	baseline	varchar(20)	string	✓	✓
SME_04	MeasureGroupId	MG1_SCE_ZON_DU000B_00005_LV_H_daysAbove	varchar(50)	string	✓	

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
SME_05	MeasureGroupPollutionReduction	null	decimal(10,2)	numeric		
SME_06	MeasureReductionAssessmentMethodId	null	varchar(100)	string		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

Measure



Description

EEA extensions

Data management rules

Attributes

Attribute details

MEA_01 – CountryCode

MEA_02 – MeasureGroupId

MEA_03 – MeasureId

MEA_04 – MeasureNationalCode

MEA_05 – MeasureName

MEA_06 – MeasureClassification

MEA_07 – MeasureType

MEA_08 – SourceSector

MEA_09 – SpatialScale

MEA_10 – ImplementationBegin

MEA_11 – ImplementationEnd

MEA_12 – MeasureCost

MEA_13 – FullEffectDate

MEA_14 – MeasureStatus

MEA_15 – ReasonIfMeasureNotUsed

MEA_16 – Deletion

Example

Description

The purpose of the Measure table is to provide information on the different measures contained in the groups of measures. It describes the spatial scale of the measures, their type and classification as well as sector of impact. The table includes also operational characteristics of measures such as: responsible entity, implementation timing, costs, status and expected time of full effect or reason if measure has not been used.

Measure (record) table has to be reported whenever a new information or an update of existing information is available. Attributes which create unique identifier of each record: CountryCode, MeasureGroupId and MeasureId.

Updates of the Measure table will be managed using reference/legacy data, where each addition and modification will be stamped with ReportingTime:

addition of new record is understood as addition of new combination of values for the attributes creating unique identifier,

modification of existing record is understood as modification of a value for any other attribute.

Validity of the new record will be tested using QC (rules to be established) - e.g. cross checking against relevant code lists and tested for coherence.

Valid modifications of existing record in this table are:

modifying ResponsibleEntity, ImplementationBegin and/or End, MeasureCost, FullEffectDate, MeasureStatus or ReasonIfMeasureNotUsed,

minor corrections e.g. of name, code or type.

Validity of the modifications of existing record will be tested using QC (rules to be established). Allowed modifications may affect outcomes of scenario analysis and/or assessments of measures, therefore re-submissions for past years, for which plans reporting has been already done, will require additional tests vs reference/legacy data before official release of such data is granted.

Data products for public will present the records validated at the latest deadline. Data products for reporters will present the records as reported recently - i.e. with the latest ReportingTime for each unique combination of CountryCode, MeasureGroupId and MeasureId, highlighting the differences between validated reports and recent reports in case of re-submissions.

Deletion of an existing record will be possible by flagging the Deletion attribute. After QCs (rules to be established), all associated records in other tables will be flagged as deleted.

EEA extensions

Measure table will be extended by several attributes:

Country (for convenience), based on ISO2 country codes,

ReportingTime, registering time of the latest reporting or modification for each record.

Data management rules

Since the unique identifier of each record consists of CountryCode, MeasureGroupId and MeasureId, the MeasureIds can be handled as static set of measures (with code, name, type, source sector) and can be re-used in different MeasureGroupIds - which can be then used for giving spatial and temporal context to the measures (spatial scale, responsible entity, implementation timing, costs, etc.).

Deletion of an existing record will be possible via the Deletion attribute (by reporting value = 1).

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
MEA_01	CountryCode	varchar(2)	string	PK	countries	
MEA_02	MeasureGroupId	varchar(50)	string	PK		ScenarioMeasure
MEA_03	MeasureId	varchar(50)	string	PK		
MEA_04	MeasureNationalCode	varchar(50)	string			
MEA_05	MeasureName	varchar(150)	string			
MEA_06	MeasureClassification	varchar(50)	string		measureclassification	
MEA_07	MeasureType	varchar(50)	string		measuretype	
MEA_08	SourceSector	varchar(50)	string		sourcesectors	

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
MEA_09	SpatialScale	varchar(50)	string		spatialscale	
MEA_10	ImplementationBegin	date	date			
MEA_11	ImplementationEnd	date	date			
MEA_12	MeasureCost	decimal(18,2)	numeric			
MEA_13	FullEffectDate	date	date			
MEA_14	MeasureStatus	varchar(50)	string		measureimplementationstatus	
MEA_15	ReasonIfMeasureNotUsed	varchar(50)	string		reasonifmeasurenotused	
MEA_16	Deletion	bit	boolean			

Note

Attributes with ReportNet3 data type `date` or `datetime` shall use the ISO 8601 format. Date-time values may include a local UTC offset.

Attribute details

MEA_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

MEA_02 – MeasureGroupId

Content

Identifier of the pollution reduction for measure group, given by data provider.

In Reference

View reference attribute

MEA_03 – MeasureId

Content

Identifier of the pollution reduction measure, given by data provider.

In Reference

View reference attribute

MEA_04 – MeasureNationalCode

Content

Unique local code assigned to the measure by the data provider.

In Reference

View reference attribute

MEA_05 – MeasureName

Content

Name or title of the pollution reduction measure.

In Reference

View reference attribute

MEA_06 – MeasureClassification

Content

Classification of the measure based on regulatory categories.

In Reference

View reference attribute

MEA_07 – MeasureType

Content

Description of the high-level implementation mechanism or scope of the measure.

In Reference

View reference attribute

MEA_08 – SourceSector

Content

Economic or activity sector targeted by the measure (e.g., transport, energy, industry).

Remarks

It can be tested against the SourceSector of SourceApportionment which was used as base for corresponding ScenarioId.

In Reference

View reference attribute

MEA_09 – SpatialScale

Content

Geographical coverage of the measure (e.g., local, national, EU-wide).

Remarks

It can be tested against the SpatialScale of SourceApportionment which was used as base for corresponding ScenarioId.

In Reference

View reference attribute

MEA_10 – ImplementationBegin

Content

Start date for implementing the measure.

In Reference

View reference attribute

MEA_11 – ImplementationEnd

Content

End date for implementing the measure.

In Reference

View reference attribute

MEA_12 – MeasureCost

Content

Estimated costs for implementing the measure over its lifetime.

In Reference

View reference attribute

MEA_13 – FullEffectDate

Content

Date when the measure is expected to reach its full impact.

In Reference

View reference attribute

MEA_14 – MeasureStatus

Content

Current status of the measure (e.g., planned, in progress, completed).

In Reference

View reference attribute

MEA_15 – ReasonIfMeasureNotUsed

Content

Explanation or justification if the measure was not implemented.

In Reference

View reference attribute

MEA_16 – Deletion

Content

Flag to indicate that this element must be deleted

Remarks

Y/N

In Reference

View reference attribute

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
MEA_01	CountryCode	DU	varchar(2)	string	✓	✓
MEA_02	MeasureGroupId	MG1_SCE_ZON_DU000B_00005_LV_H_daysAbove	varchar(50)	string	✓	
MEA_03	MeasureId	Measure_1256	varchar(50)	string	✓	
MEA_04	MeasureNationalCode	Measure_1256	varchar(50)	string		
MEA_05	MeasureName	Industrial Emission Control	varchar(150)	string		
MEA_06	MeasureClassification	emissioncontrol	varchar(50)	string		✓
MEA_07	MeasureType	outside	varchar(50)	string		✓
MEA_08	SourceSector	industry	varchar(50)	string		✓
MEA_09	SpatialScale	local	varchar(50)	string		✓

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
MEA_10	ImplementationBegin	2025-02-01	date	date		
MEA_11	ImplementationEnd	2027-12-31	date	date		
MEA_12	MeasureCost	null	decimal(18,2)	numeric		
MEA_13	FullEffectDate	null	date	date		
MEA_14	MeasureStatus	implementation	varchar(50)	string		✓
MEA_15	ReasonIfMeasureNotUsed	null	varchar(50)	string		✓
MEA_16	Deletion	0	bit	boolean		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

Documentation



Description

EEA extensions

Data management rules

Attributes

Attribute details

DOC_01 – CountryCode

DOC_02 – DataTable

DOC_03 – DocumentType

DOC_04 – DocumentId

DOC_05 – DocumentAttachment

DOC_06 – DocumentOriginalURL

Example

Description

The purpose of the Documentation table is to communicate the documents requested in different tables such as documentation on data quality process, documentation on the models, the air quality plans etc.

Documentation (record) table has to be reported whenever a new document or an updated document is available. Attributes which create unique identifier of each record: CountryCode, DataTable, DocumentType and DocumentId.

Updates of the Documentation table will be managed using reference/legacy data, where each addition and modification will be stamped with the ReportingTime. Data products for public and reporters will present the records with the latest ReportingTime for each unique identifier.

EEA extensions

Documentation table will be extended by EEA by:

Country (for convenience), based on ISO2 country codes;

ReportingTime, registering time of the latest reporting or modification for each record;

DocumentURL where the document is stored.

Data management rules

More details about rules of data management related to documentation reporting are given above under the relevant tables.

Records in Document table which do not have any correspondence (via DocumentId) to other tables will be flagged for deletion in the reference data flow.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
DOC_01	CountryCode	varchar(2)	string	PK	countries	
DOC_02	DataTable	varchar(50)	string	PK	datatable	
DOC_03	DocumentType	varchar(50)	string	PK	documenttype	
DOC_04	DocumentId	varchar(500)	string	PK		MeasurementStation SamplingProcess ModelObjectiveEstimation AssessmentRegimeZone CompliancePlanLink SourceApportionment PollutionLevelAdjustment
DOC_05	DocumentAttachment	varchar(100)	attachment (R3 data type)			
DOC_06	DocumentOriginalURL	varchar(100)	string			

Attribute details

DOC_01 – CountryCode

Content

Country or territory ISO2 code.

In Reference

View reference attribute

DOC_02 – DataTable

Content

Table name of origin.

In Reference

View reference attribute

DOC_03 – DocumentType

Content

Document content type.

In Reference

View reference attribute

DOC_04 – DocumentId

Content

Identifier of the document given by data provider.

In Reference

View reference attribute

DOC_05 – DocumentAttachment

Content

Attached document PDF.

DOC_06 – DocumentOriginalURL

Content

URL of the attached document PDF.

In Reference

View reference attribute

Example

Attribute Code	Attribute Name	Example	SQL DB Data Type	ReportNet3 Data Type	PK	CL
DOC_01	CountryCode	DU	varchar(2)	string	✓	✓
DOC_02	DataTable	Station	varchar(50)	string	✓	✓
DOC_03	DocumentType	NetworkDocument	varchar(50)	string	✓	✓
DOC_04	DocumentId	DOC_NET_DU01	varchar(500)	string	✓	
DOC_05	DocumentAttachment	DUS_NET_DU01.pdf	varchar(100)	attachment (R3 data type)		
DOC_06	DocumentOriginalURL	www.dea.dustovia.dus/AQeREP/NET/DU01.pdf	varchar(100)	string		

Legend: PK = Primary Key; CL = Code List.

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

ObservationMeasurementResultPNSD



Description

EEA extensions

Data management rules

Attributes

Attribute details

OMP_01 – CountryCode

OMP_02 – AssessmentMethodId

OMP_03 – Start

OMP_04 – LowerRange

OMP_05 – PollutantId

OMP_06 – End

OMP_07 – Value

OMP_08 – Unit

OMP_09 – Validity

OMP_10 – Verification

OMP_11 – TimeResolution

OMP_12 – ResultTime

OMP_13 – UpperRange

OMP_14 – Temperature

OMP_15 – RelativeHumidity

OMP_16 – Pressure

OMP_17 – Inversion

Example

Description

The purpose of the ObservationMeasurementResult table is to collect information on air quality measurement results. It describes the AQ values and time reference for each value. Location of each measurement can be identified through the SamplingPoint table.

ObservationMeasurementResult table has to be reported every year because new information is expected at each reporting cycle. Attributes which create unique identifier of each record: CountryCode, AssessmentMethodId and Start.

Updates of the ObservationMeasurementResult table will not be time stamped - i.e. each update generates the only version of the time series with AQ values that is stored at the EEA, as well as the only version of the AQ statistics stored in the reference/legacy data:

either as addition of a new record, understood as addition of a new combination of values for the attributes creating the unique identifier;

or modification of an existing record, understood as modification of a value for any other attribute.

Validity of new records will be tested using QC procedures (rules to be established), including cross-checking against existing records in the SamplingPoint table.

Valid modifications of existing records are based on re-submission of time series for the existing set of attributes creating the unique identifier:

either to overwrite or correct values of other attributes;

or to modify the Validity flag to `-1` for deletion purposes.

Validity of modifications will be tested through QC procedures. Allowed modifications may affect outcomes of compliance analysis and/or assessments. Therefore, re-submissions for past years, for which compliance reporting has already been completed, will require additional tests against reference/legacy data before official release of such data is granted.

Data products for the public (download services for time series and aggregated values) will present records validated at the latest deadline. Data products for reporters will present records as most recently reported, highlighting differences between validated reports and recent reports in case of re-submissions.

EEA extensions

Reference data flow will contain two tables representing reporting of time series:

ObservationMeasurementInventory;

ObservationMeasurementStatistics.

ObservationMeasurementInventory will collect information about counts of values reported for each sampling point and year with specific settings (validity, verification, time resolution, etc.) and will contain a reference to the raw data file. ObservationMeasurementStatistics in the reference/legacy data flow will differ significantly from the reporting table because it will only contain aggregated values.

The scope of aggregations stored in that table remains to be agreed (P1D ?, P1M ?, annual only ?, etc.). SamplingPointReferenceId will be used in the reference/legacy data flow together with AssessmentMethodId in order to preserve continuity.

ObservationMeasurementStatistics will also be extended by attributes such as:

Pollutant (for convenience), based on code list;

DataCoverage, as a feature of aggregated data;

DataAggregationProcessId (P1Y, etc.);

DataAggregationProcess, based on DataAggregationProcessId;

SourceDataFlow (AirBase, R2 e-Reporting, R3 e-Reporting, UTD).

Data management rules

Deletion of records from ObservationMeasurementResult can be achieved by re-submitting the same records with the Validity flag set to `-1`.

Attributes

Attribute Code	Attribute Name	SQL DB Data Type	ReportNet3 Data Type	Properties	Code list	Related table(s)
OMP_01	CountryCode	varchar(2)	string	PK	countries	
OMP_02	AssessmentMethodId	varchar(100)	string	PK		
OMP_03	Start	datetime	datetime	PK		
OMP_04	LowerRange	int	numeric	PK		
OMP_05	PollutantId	int	numeric		pollutant	
OMP_06	End	datetime	datetime			
OMP_07	Value	decimal(10,2)	numeric			
OMP_08	Unit	varchar(10)	string		concentrationunit	
OMP_09	Validity	int	numeric		observationvalidity	
OMP_10	Verification	int	numeric		observationverification	
OMP_11	TimeResolution	varchar(10)	string		primaryObservation	
OMP_12	ResultTime	datetime	datetime			
OMP_13	UpperRange	int	numeric			
OMP_14	Temperature	decimal(5,2)	numeric			
OMP_15	RelativeHumidity	decimal(5,2)	numeric			
OMP_16	Pressure	decimal(5,2)	numeric			
OMP_17	Inversion	varchar(20)	string		PNSDinversion	

Note

Attributes with ReportNet3 data type `date` or `datetime` shall use the ISO 8601 format. Date-time values may include a local UTC offset.

Attribute details

OMP_01 – CountryCode

Content

Country or territory ISO2 code.

OMP_02 – AssessmentMethodId

Content

Identifier of the assessment method (sampling point), given by data provider.

Remarks

It will be cross-checked against the SamplingPoint table.

OMP_03 – Start

Content

Start date and time of the measurement period.

Remarks

Start must be provided with time zone information.

OMP_04 – LowerRange

Content

Lower range of PNSD bin.

OMP_05 – PollutantId

Content

Code of the air pollutant being measured, as per Data Dictionary standards.

Remarks

End must be provided with time zone information.

OMP_06 – End

Content

End date and time of the measurement period.

OMP_07 – Value

Content

Measured concentration or level of the air pollutant.

OMP_08 – Unit

Content

Unit of measurement for the air pollution level (e.g., µg/m³).

OMP_09 – Validity

Content

Indicator of whether the measurement data is valid or not.

OMP_10 – Verification

Content

Information based on verification flags found in reported time series.

OMP_11 – TimeResolution

Content

Time resolution of the reported measurement observations (e.g., hourly, daily).

OMP_12 – ResultTime

Content

Time at which the result was generated or recorded.

OMP_13 – UpperRange

Content

Upper range of PNSD bin.

OMP_14 – Temperature

Content

Temperature measured along the PNSD (deg Celsius).

OMP_15 – RelativeHumidity

Content

Relative humidity measured along the PNSD (%).

OMP_16 – Pressure

Content

Pressure measured along the PNSD (hPa).

OMP_17 – Inversion

Content

Inversion method used by country (code list flag).

Example

Note: The example file is temporarily unavailable while it is being updated. The updated version will be made available once the revision is complete.

Identifiers

Description

Identifier categories

Fully mandatory identifier formats

Partially mandatory identifier formats

Free-format identifiers

Separators

AssessmentRegimeId

SamplingPointReferenceId

AttainmentId

ScenarioId

AssessmentMethodId for models and OBE

PlanId

Description

This section describes the identifier formats used throughout the reporting schema.

Some identifiers follow a fully mandatory naming convention in order to ensure consistency across reporting authorities and facilitate automatic processing by the EEA. Other identifiers are partially constrained or completely free.

Unless specified otherwise, identifier values must be unique within the scope of the reporting country.

Identifier categories

Fully mandatory identifier formats

The following identifiers must follow the format specified in this guide:

AssessmentRegimeId

SamplingPointReferenceId

AttainmentId

ScenarioId

Partially mandatory identifier formats

The following identifiers have mandatory prefixes or structural elements, while other parts may be defined by the reporting authority:

AssessmentMethodId (models and OBE)

PlanId

Free-format identifiers

All remaining identifiers may be freely defined by reporting authorities.

For readability and easier interpretation, it is strongly recommended to use meaningful prefixes indicating the identifier type, for example:

Prefix	Meaning
NET	Network
DOC	Document
SAP	Source Apportionment
SRS	Spatial Representativeness
EXC	Exceedance Extent
PLA	Plan
SCE	Scenario
MOD	Model
OBE	Objective Estimation

Separators

For fully mandatory identifiers, the separator must be:

```
_
```

For all other identifiers, including partially mandatory identifiers, any separator may be used, for example:

```
_
-
.
```

AssessmentRegimeId

Referenced in: ARZ_02

Mandatory format:

```
ARE_<ZoneId>_<PollutantId>_<ObjectiveType>_<ProtectionTarget>_<ReportingMetric>_<ClassificationYear>_<Index>
```

Example:

```
ARE_ZON_DU000A_0005_LV_H_aMean_2021_1
```

Element	Description
ARE	Fixed prefix indicating Assessment Regime
ZoneId	Air quality zone identifier
PollutantId	Pollutant code
ObjectiveType	Environmental objective type
ProtectionTarget	Protection target
ReportingMetric	Reporting metric
ClassificationYear	Year of classification
Index	Sequential number

SamplingPointReferenceId

Referenced in: SPO_03

Mandatory format:

```
SPOref_<StationEoICode>_<PollutantId>_<Index>
```

Example:

```
SPOref_DU0001_0005_1
```

Element	Description
SPOref	Fixed prefix indicating Sampling Point Reference
StationEoICode	Station identifier
PollutantId	Pollutant code
Index	Sequential number

AttainmentId

Referenced in: CAM_15

Mandatory format:

```
ATT_<ZoneId>_<PollutantId>_<ObjectiveType>_<ProtectionTarget>_<ReportingMetric>_<ReportingYear>_<Index>
```

Example:

```
ATT_ZON_DU000A_00005_LV_H_aMean_2024_1
```

Element	Description
ATT	Fixed prefix indicating compliance situation
ZoneId	Air quality zone identifier
PollutantId	Pollutant code
ObjectiveType	Environmental objective type
ProtectionTarget	Protection target
ReportingMetric	Reporting metric
ReportingYear	Reporting year
Index	Sequential number

ScenarioId

Referenced in: PSC_03, CPL_04

Mandatory format:

```
SCE_<ZoneId>_<PollutantId>_<ObjectiveType>_<ProtectionTarget>_<ReportingMetric>_<Index>
```

Example:

```
SCE_ZON_DU000B_00005_LV_H_daysAbove_1
```

Element	Description
SCE	Fixed prefix indicating scenario
ZoneId	Air quality zone identifier
PollutantId	Pollutant code
ObjectiveType	Environmental objective type
ProtectionTarget	Protection target
ReportingMetric	Reporting metric
Index	Sequential number

AssessmentMethodId for models and OBE

Referenced in: MOE_02

Partially mandatory format.

Recommended prefixes:

```
MOD_
OBE_
```

Examples:

```
MOD_DU_PM10_LOC_SCEN_REF
OBE_DU_SEASALT_PM10_H_LV_ADJ
```

Prefix	Meaning
MOD	Modelling application
OBE	Objective estimation

PlanId

Referenced in: CPL_03

Partially mandatory format.

Recommended prefix:

```
PLA_
```

Example:

```
PLA_ZON_DU000B_00005_LV_H_daysAbove
```

Element	Description
PLA	Fixed prefix indicating plan
Remaining elements	Defined by reporting authority

Table Relationships

Purpose

This document describes the relationships between tables in the AQ eReporting reporting schema.

The relationships are divided into two categories:

Physical primary key–foreign key relationships

Logical relationships that cannot be represented correctly by a conventional SQL foreign key

The classification is made per relationship rather than per table.

A table may therefore participate in:

a physical foreign key relationship with one table; and

a logical relationship with another table.

Relationship Categories

Physical PK–FK Relationships

A physical foreign key can be defined where:

the parent columns form the complete documented primary key;

the child table contains all columns of the parent key;

the relationship points to one uniquely identifiable parent row;

the relationship applies unconditionally;

the relationship is based on equality between columns;

the relationship does not depend on dates, types, categories, shared datasets or alternative parent tables.

Logical Relationships

A relationship remains logical where:

the child contains only part of the parent primary key;

the child refers to a group of parent records;

one result dataset is shared by several parent records;

the target parent table depends on a type or discriminator;

the relationship depends on a validity period;

the relationship uses a business identifier that is not the complete parent key;

the relationship requires additional context;

the relationship applies only under certain conditions;

the relationship cannot be represented correctly through a standard equality-based foreign key.

Physical PK–FK Relationships

MeasurementStation ← SamplingPoint

Relationship Description

Every **SamplingPoint** belongs to one **MeasurementStation**.

The parent station is uniquely identified by:

`CountryCode`

`StationEoICode`

The child contains the complete documented primary key of the parent.

MeasurementStation Primary Key

```
ALTER TABLE [reporting].[MeasurementStation]
ADD CONSTRAINT [PK_MeasurementStation]
PRIMARY KEY
(
    [CountryCode],
    [StationEoICode]
);
```

SamplingPoint Primary Key

```
ALTER TABLE [reporting].[SamplingPoint]
ADD CONSTRAINT [PK_SamplingPoint]
PRIMARY KEY
(
    [CountryCode],
    [AssessmentMethodId]
);
```

Foreign Key

```
ALTER TABLE [reporting].[SamplingPoint]
ADD CONSTRAINT [FK_SamplingPoint_MeasurementStation]
FOREIGN KEY
(
    [CountryCode],
    [StationEoICode]
)
REFERENCES [reporting].[MeasurementStation]
(
    [CountryCode],
    [StationEoICode]
);
```

SamplingPoint ← SamplingPointLocation

Relationship Description

Every **SamplingPointLocation** record belongs to one **SamplingPoint**.

`LocationBegin` distinguishes different location-validity periods for the same sampling point.

It is part of the child primary key, but it is not required to identify the parent sampling point.

A sampling point may have:

several different sampling point locations;
several successive validity periods for the same location.

SamplingPointLocation Primary Key

```
ALTER TABLE [reporting].[SamplingPointLocation]
ADD CONSTRAINT [PK_SamplingPointLocation]
PRIMARY KEY
(
    [CountryCode],
    [AssessmentMethodId],
    [LocationBegin]
);
```

Foreign Key

```
ALTER TABLE [reporting].[SamplingPointLocation]
ADD CONSTRAINT [FK_SamplingPointLocation_SamplingPoint]
FOREIGN KEY
(
    [CountryCode],
    [AssessmentMethodId]
)
REFERENCES [reporting].[SamplingPoint]
(
    [CountryCode],
    [AssessmentMethodId]
);
```

SamplingPoint ← SamplingProcess

Relationship Description

Every **SamplingProcess** record belongs to one **SamplingPoint**.

A sampling point may have:

several different sampling processes;
several successive validity periods for the same process.

SamplingProcess Primary Key

```
ALTER TABLE [reporting].[SamplingProcess]
ADD CONSTRAINT [PK_SamplingProcess]
PRIMARY KEY
(
    [CountryCode],
    [ProcessId],
    [AssessmentMethodId],
    [ProcessActivityBegin]
);
```

Foreign Key

```
ALTER TABLE [reporting].[SamplingProcess]
ADD CONSTRAINT [FK_SamplingProcess_SamplingPoint]
FOREIGN KEY
(
    [CountryCode],
    [AssessmentMethodId]
)
REFERENCES [reporting].[SamplingPoint]
(
    [CountryCode],
    [AssessmentMethodId]
);
```

SamplingPoint ← ObservationMeasurementResult

Relationship Description

Every observation result belongs to an existing **SamplingPoint**.

The physical foreign key confirms that the sampling point exists.

ObservationMeasurementResult Primary Key

```
ALTER TABLE [reporting].[ObservationMeasurementResult]
ADD CONSTRAINT [PK_ObservationMeasurementResult]
PRIMARY KEY
(
    [CountryCode],
    [AssessmentMethodId],
    [Start]
);
```

Foreign Key

```
ALTER TABLE [reporting].[ObservationMeasurementResult]
ADD CONSTRAINT [FK_ObservationMeasurementResult_SamplingPoint]
FOREIGN KEY
(
    [CountryCode],
    [AssessmentMethodId]
)
REFERENCES [reporting].[SamplingPoint]
(
    [CountryCode],
    [AssessmentMethodId]
);
```

Additional Logical Relationships

The following observation relationships remain logical:

```
ObservationMeasurementResult
    ↔ SamplingPointLocation through the observation validity period

ObservationMeasurementResult
    ↔ SamplingProcess through the observation validity period
```

The correct location and process records are determined from the observation timestamp.

SamplingPoint ← ObservationMeasurementResultPNSD

Relationship Description

Every **PNSD observation** belongs to one existing **SamplingPoint**.

The combination:

`CountryCode`
`AssessmentMethodId`
`Start`
`LowerRange`
uniquely identifies an individual PNSD result row.

`LowerRange` distinguishes different particle-size intervals reported for the same sampling point and observation start time.

It is part of the child primary key but is not needed to identify the parent **SamplingPoint**.

The relationship to **SamplingPoint** therefore remains a proper physical PK–FK relationship.

ObservationMeasurementResultPNSD Primary Key

The documented primary key includes `LowerRange`.

```
ALTER TABLE [reporting].[ObservationMeasurementResultPNSD]
ADD CONSTRAINT [PK_ObservationMeasurementResultPNSD]
PRIMARY KEY
(
    [CountryCode],
    [AssessmentMethodId],
    [Start],
    [LowerRange]
);
```

Foreign Key

```
ALTER TABLE [reporting].[ObservationMeasurementResultPNSD]
ADD CONSTRAINT [FK_ObservationMeasurementResultPNSD_SamplingPoint]
FOREIGN KEY
(
    [CountryCode],
    [AssessmentMethodId]
)
REFERENCES [reporting].[SamplingPoint]
(
    [CountryCode],
    [AssessmentMethodId]
);
```

Additional Logical Relationships

The following PNSD relationships remain logical:

```
ObservationMeasurementResultPNSD
    ↔ SamplingPointLocation through the observation validity period

ObservationMeasurementResultPNSD
    ↔ SamplingProcess through the observation validity period
```

The correct location and process records are determined from the PNSD observation timestamp.

ModelObjectiveEstimation ← MOEResultInline

Relationship Description

Every inline model grid-cell result belongs to one model or objective-estimation definition and one aggregation process.

The following child columns identify the individual result cell:

`Start`
`X`
`Y`

They are not required to identify the parent method.

Validation that the parent record uses `inline` result encoding remains a separate logical consistency rule.

ModelObjectiveEstimation Primary Key

```
ALTER TABLE [reporting].[ModelObjectiveEstimation]
ADD CONSTRAINT [PK_ModelObjectiveEstimation]
PRIMARY KEY
(
    [CountryCode],
    [AssessmentMethodId],
    [DataAggregationProcessId]
);
```

MOEResultInline Primary Key

```
ALTER TABLE [reporting].[MOEResultInline]
ADD CONSTRAINT [PK_MOEResultInline]
PRIMARY KEY
(
    [CountryCode],
    [AssessmentMethodId],
    [Start],
    [DataAggregationProcessId],
    [X],
    [Y]
);
```

Foreign Key

```
ALTER TABLE [reporting].[MOEResultInline]
ADD CONSTRAINT [FK_MOEResultInline_ModelObjectiveEstimation]
FOREIGN KEY
(
    [CountryCode],
    [AssessmentMethodId],
    [DataAggregationProcessId]
)
REFERENCES [reporting].[ModelObjectiveEstimation]
(
    [CountryCode],
    [AssessmentMethodId],
    [DataAggregationProcessId]
);
```

ModelObjectiveEstimation ← MOEResultExternal

Relationship Description

Every external model result belongs to one model or objective-estimation definition and one aggregation process.

The physical FK establishes that the model definition exists.

Validation that the parent record uses **‘external’** result encoding remains a separate logical consistency rule.

MOEResultExternal Primary Key

```
ALTER TABLE [reporting].[MOEResultExternal]
ADD CONSTRAINT [PK_MOEResultExternal]
PRIMARY KEY
(
    [CountryCode],
    [AssessmentMethodId],
    [Start],
    [DataAggregationProcessId]
);
```

Foreign Key

```
ALTER TABLE [reporting].[MOEResultExternal]
ADD CONSTRAINT [FK_MOEResultExternal_ModelObjectiveEstimation]
FOREIGN KEY
(
    [CountryCode],
    [AssessmentMethodId],
    [DataAggregationProcessId]
)
REFERENCES [reporting].[ModelObjectiveEstimation]
(
    [CountryCode],
    [AssessmentMethodId],
    [DataAggregationProcessId]
);
```

AssessmentRegimeZone ← ComplianceAssessmentMethod

Relationship Description

Every **ComplianceAssessmentMethod** record belongs to one assessment regime.

The child contains the complete documented PK of **AssessmentRegimeZone**.

AssessmentRegimeZone Primary Key

```
ALTER TABLE [reporting].[AssessmentRegimeZone]
ADD CONSTRAINT [PK_AssessmentRegimeZone]
PRIMARY KEY
(
    [CountryCode],
    [AssessmentRegimeId]
);
```

ComplianceAssessmentMethod Primary Key

```
ALTER TABLE [reporting].[ComplianceAssessmentMethod]
ADD CONSTRAINT [PK_ComplianceAssessmentMethod]
PRIMARY KEY
(
    [CountryCode],
    [AttainmentId],
    [DataAggregationProcessId],
    [AssessmentMethodId]
);
```

Foreign Key

```
ALTER TABLE [reporting].[ComplianceAssessmentMethod]
ADD CONSTRAINT [FK_ComplianceAssessmentMethod_AssessmentRegimeZone]
FOREIGN KEY
(
    [CountryCode],
    [AssessmentRegimeId]
)
REFERENCES [reporting].[AssessmentRegimeZone]
(
    [CountryCode],
    [AssessmentRegimeId]
);
```

ModelObjectiveEstimation ← PlanScenario

Relationship Description

`ScenarioAssessmentMethodId` identifies the model or objective-estimation method used for the scenario and scenario category within a plan.

Together with:

`CountryCode`

`DataAggregationProcessId`

it resolves to the complete **ModelObjectiveEstimation** primary key.

The corresponding parent and child column names do not need to be identical, but:

their meaning must correspond;
their SQL data types must be compatible;
their lengths must be compatible.
If `ScenarioAssessmentMethodId` is nullable, SQL Server permits a null FK value.

Whether the model reference is mandatory must therefore be enforced through nullability or separate validation.

PlanScenario Primary Key

```
ALTER TABLE [reporting].[PlanScenario]
ADD CONSTRAINT [PK_PlanScenario]
PRIMARY KEY
(
    [CountryCode],
    [PlanId],
    [ScenarioId],
    [ScenarioCategory]
);
```

Foreign Key

```
ALTER TABLE [reporting].[PlanScenario]
ADD CONSTRAINT [FK_PlanScenario_ModelObjectiveEstimation]
FOREIGN KEY
(
    [CountryCode],
    [ScenarioAssessmentMethodId],
    [DataAggregationProcessId]
)
REFERENCES [reporting].[ModelObjectiveEstimation]
(
    [CountryCode],
    [AssessmentMethodId],
    [DataAggregationProcessId]
);
```

Physical Relationship Summary

The following relationships can be represented by ordinary physical PK–FK constraints, subject to compatible SQL data types and valid existing data.

Physical PK–FK Relationship List

MeasurementStation

← SamplingPoint

SamplingPoint

← SamplingPointLocation
← SamplingProcess
← ObservationMeasurementResult
← ObservationMeasurementResultPNSD

ModelObjectiveEstimation

← MOEResultInline
← MOEResultExternal
← PlanScenario

AssessmentRegimeZone

← ComplianceAssessmentMethod

Logical Relationships

Authority → Multiple Entity Tables

Relationship

`Authority.AuthorityInstanceId` can identify different types of reporting entities.

Depending on the value of `AuthorityInstance`, or an equivalent type attribute, it may refer to:

`MeasurementStation.StationEoICode`
`SamplingPoint.AssessmentMethodId`
`AssessmentRegimeZone.ZoneId`
`ZoneGeometry.ZoneId`
`ModelObjectiveEstimation.AssessmentMethodId`
or to a national level if `AuthorityInstance = 'nuts0'`.

Why No Physical FK Can Be Defined

The target parent table depends on another attribute.

A single SQL foreign key cannot conditionally reference different parent tables.

Creating several foreign keys would incorrectly require the same identifier to exist in **all** possible parent tables.

This is therefore a **polymorphic logical relationship**.

SamplingPointLocation ↔ ObservationMeasurementResult

Relationship

An observation must correspond to a valid sampling-point location for the observation time.

The matching logic uses:

same `CountryCode`
same `AssessmentMethodId`
`Observation.Start >= LocationBegin`
`Observation.Start < LocationEnd`
or:

`LocationEnd` is `NULL`

Why No Physical FK Can Be Defined

The relationship depends on the observation timestamp falling within a validity interval.

`ObservationMeasurementResult` does not contain `LocationBegin`, which forms part of the `SamplingPointLocation` primary key.

It therefore does not directly identify one specific location record.

This is a **temporal logical relationship**.

SamplingProcess ↔ ObservationMeasurementResult

Relationship

An observation must correspond to an active sampling process for the same sampling point.

The matching logic uses:

same `CountryCode`
same `AssessmentMethodId`
`Observation.Start >= ProcessActivityBegin`
`Observation.Start < ProcessActivityEnd`
or:

`ProcessActivityEnd` is `NULL`

Why No Physical FK Can Be Defined

The `SamplingProcess` primary key contains:

`CountryCode`
`ProcessId`
`AssessmentMethodId`
`ProcessActivityBegin`
The observation does not identify:

`ProcessId`
`ProcessActivityBegin`
The applicable process must therefore be determined from the observation time and the process validity period.

SamplingPointLocation and SamplingProcess ↔ ObservationMeasurementResultPNSD

Relationship

Similarly as observations, the PNSD observations must correspond to:

an existing `SamplingPoint`;
a valid `SamplingPointLocation`;
an active `SamplingProcess`.
The relationship to `SamplingPoint` can be implemented through a physical FK.

The location and process relationships remain logical.

Why No Physical FK Can Be Defined

Similarly as observations, the PNSD observations do not contain the complete primary key of the applicable location or process record.

The correct record is selected using the PNSD observation timestamp.

AssessmentRegimeZone → ZoneGeometry

Relationship

For assessment regimes represented by a reported AQ zone:

`AssessmentRegimeZone`

```
(CountryCode, ZoneId)
```

uses

`ZoneGeometry`

```
(CountryCode, ZoneId)
```

Why No Physical FK Can Be Defined

The equality columns correspond to the documented `ZoneGeometry` key.

However, a geometry record may not be required for every possible zone reference. An assessment-regime record may instead identify an area through a NUTS reference (`ZoneCategory = 'nuts'`).

A mandatory physical FK would reject such valid records.

The relationship is therefore conditional on the zone type or zone category.

The data types and lengths of `ZoneId` must also be compatible before a physical relationship could be considered.

ComplianceAssessmentMethod → SamplingPoint or ModelObjectiveEstimation

Relationship

`ComplianceAssessmentMethod.AssessmentMethodId` may identify either:

`SamplingPoint.AssessmentMethodId`
or

`ModelObjectiveEstimation.AssessmentMethodId`
For a model or objective-estimation method, `DataAggregationProcessId` is also relevant.

Why No Physical FK Can Be Defined

This is an **alternative-parent relationship**.

A foreign key to `SamplingPoint` would reject model records.

A foreign key to `ModelObjectiveEstimation` would reject sampling-point records.

Applying both would incorrectly require every assessment method to exist in both tables.

The correct target table must be selected according to the assessment-method type.

ComplianceAssessmentMethod → SpatialRepresentativeness

Relationship

`ComplianceAssessmentMethod`

```
(CountryCode, SRSId)
```

uses one or more

`SpatialRepresentativeness`

```
(CountryCode, SRSId, SRSApplicationId)
```

One `SRSId` may have several applications.

Why No Physical FK Can Be Defined

`ComplianceAssessmentMethod` does not contain `SRSApplicationId`, which forms part of the `SpatialRepresentativeness` primary key.

The compliance record intentionally points to the complete SRS group rather than to one particular SRS application, as it may be related both:

to the spatial representativeness area of a sampling point;
to the exceedance extent (when the compliance record describes an exceedance).

SpatialRepresentativeness ↔ SRSInline

Relationship

`SRSInline` stores a potentially large inline spatial result dataset identified by:

`CountryCode + SRSApplicationId`

The same application dataset may be used by more than one `SpatialRepresentativeness` record.

Example:

SpatialRepresentativeness

```
CountryCode | SRSId   | SRSApplicationId
------------+---------+-----------------
DK          | SRS-001 | APP-01
DK          | SRS-002 | APP-01
```

The shared geometry or grid dataset is stored once:

SRSInline

```
CountryCode | SRSApplicationId | X  | Y
------------+------------------+----+----
DK          | APP-01           | X1 | Y1
DK          | APP-01           | X2 | Y2
DK          | APP-01           | X3 | Y3
...
```

All rows belonging to application `APP-01` are used by both SRS records.

Why No Physical FK Can Be Defined

The `SpatialRepresentativeness` primary key is:

`CountryCode`
`SRSId`
`SRSApplicationId`
`SRSInline` contains only:

`CountryCode`
`SRSApplicationId`
The shorter combination in `SpatialRepresentativeness` is intentionally not necessarily unique.

The same application dataset may belong to several `SRSId` values.

Repeating `SRSId` in every `SRSInline` geometry or grid-cell row would:

duplicate the value over potentially thousands of rows;
increase the amount of submitted data;
incorrectly represent a shared application dataset as separate duplicated datasets.
The logical rule is:

Every `SRSInline` application dataset must be used by at least one `SpatialRepresentativeness` record with the same `CountryCode` and `SRSApplicationId`.
Validation that the corresponding `SpatialRepresentativeness` record uses **'inline'** result encoding remains a separate logical consistency rule.

SpatialRepresentativeness ↔ SRSExternal

Relationship

`SRSExternal` stores an external spatial dataset identified by:

`CountryCode + SRSApplicationId`

The external result may be used by one or more `SpatialRepresentativeness` records.

Why No Physical FK Can Be Defined

`SRSExternal` does not contain `SRSId`, while the complete `SpatialRepresentativeness` primary key does.

The combination:

`CountryCode + SRSApplicationId`

may legitimately match several `SpatialRepresentativeness` records.

The external result dataset is deliberately stored once and shared.

Validation that the corresponding `SpatialRepresentativeness` record uses **'external'** result encoding remains a separate logical consistency rule.

SpatialRepresentativeness → ModelObjectiveEstimation

Relationship

`SpatialRepresentativeness.RepresentativenessAssessmentMethodId`

identifies

`ModelObjectiveEstimation.AssessmentMethodId`

The model or objective-estimation method is used for generating the spatial-representativeness result.

Why No Physical FK Can Be Defined

The complete `ModelObjectiveEstimation` primary key also contains:

`DataAggregationProcessId`
`SpatialRepresentativeness` does not contain the complete parent key.

The relationship identifies the assessment method generally, rather than one uniquely identified method-and-aggregation row.

Derivation and Use of DataAggregationProcessId

The `DataAggregationProcessId` corresponding to `SpatialRepresentativeness.RepresentativenessAssessmentMethodId` may be derived indirectly through the associated `SRSId` in the `ComplianceAssessmentMethod` table.

The relationship can be interpreted as follows:

```
SpatialRepresentativeness.SRSId
    → ComplianceAssessmentMethod.SRSId

ComplianceAssessmentMethod.DataAggregationProcessId
    → ModelObjectiveEstimation.DataAggregationProcessId
```

This derived value may be useful for comparison with the `DataAggregationProcessId` of the corresponding `AssessmentMethodId` in the `ModelObjectiveEstimation` table.

However, this comparison should **not** be interpreted as a mandatory equality rule.

A spatial-representativeness area produced using modelling results for one aggregation type (for example an annual mean) may also be applicable to other aggregation types.

Consequently, the derived `DataAggregationProcessId` should be treated primarily as contextual or consistency information rather than as a required component of a physical PK–FK relationship or a strict validation condition.

ComplianceAssessmentMethod ↔ PollutionLevelAdjustment

Relationship

`PollutionLevelAdjustment`

```
(CountryCode, AttainmentId)
```

corresponds to

`ComplianceAssessmentMethod`

```
(CountryCode, AttainmentId)
```

The adjustment belongs to the compliance or attainment situation represented by `AttainmentId`.

Why No Physical FK Can Be Defined

`AttainmentId` is not the complete primary key of `ComplianceAssessmentMethod`.

The CAM primary key also includes:

`DataAggregationProcessId`

`AssessmentMethodId`

The adjustment refers to the attainment context rather than to one complete CAM row.

PollutionLevelAdjustment → ModelObjectiveEstimation

Relationship

`PollutionLevelAdjustment.AdjustmentAssessmentMethodId`

identifies

`ModelObjectiveEstimation.AssessmentMethodId`

The referenced model or objective-estimation method is used to calculate the adjustment.

Why No Physical FK Can Be Defined

The adjustment table does not contain:

`DataAggregationProcessId`

which forms part of the complete `ModelObjectiveEstimation` primary key.

The relationship is therefore at assessment-method level rather than at one specific model-and-aggregation record.

Derivation and Use of DataAggregationProcessId

The `DataAggregationProcessId` corresponding to `PollutionLevelAdjustment.AdjustmentAssessmentMethodId` may be derived indirectly through the associated `AttainmentId` in the `ComplianceAssessmentMethod` table.

The relationship can be interpreted as follows:

```
PollutionLevelAdjustment.AttainmentId
    → ComplianceAssessmentMethod.AttainmentId

ComplianceAssessmentMethod.DataAggregationProcessId
    → ModelObjectiveEstimation.DataAggregationProcessId
```

This derived value may be useful for comparison with the `DataAggregationProcessId` of the corresponding `AssessmentMethodId` in the `ModelObjectiveEstimation` table.

However, this comparison should **not** be interpreted as a mandatory equality rule.

The temporal resolution of the modelling results used to calculate an adjustment is expected to be finer than the target aggregation type to which the adjustment is applied.

For example, daily adjustment values may be used to adjust an annual mean or to calculate an adjusted number of exceedance days.

Consequently, the derived `DataAggregationProcessId` should be treated primarily as contextual or comparison information rather than as a required component of a physical PK–FK relationship or a strict validation condition.

ComplianceAssessmentMethod ↔ CompliancePlanLink

Relationship

`CompliancePlanLink`

```
(CountryCode, AttainmentId)
```

corresponds to

`ComplianceAssessmentMethod`

```
(CountryCode, AttainmentId)
```

The plan link associates an attainment situation with:

a plan;
a scenario;
a source-apportionment dataset.

Why No Physical FK Can Be Defined

`AttainmentId` is not the complete `ComplianceAssessmentMethod` primary key.

The relationship points to the attainment context, which may be represented by a wider set of CAM records (for example several assessment methods indicating exceedance under the same `AttainmentId`).

CompliancePlanLink → PlanScenario

Relationship

`CompliancePlanLink`

`(CountryCode, PlanId, ScenarioId)`

corresponds to

`PlanScenario`

`(CountryCode, PlanId, ScenarioId, ScenarioCategory)`

Why No Physical FK Can Be Defined

`CompliancePlanLink` does not contain `ScenarioCategory`, which forms part of the `PlanScenario` primary key.

The link points to the plan-and-scenario context without identifying one scenario-category row.

In other words, it points to the whole scenario-category set for each scenario.

CompliancePlanLink → SourceApportionment

Relationship

`CompliancePlanLink`

`(CountryCode, SourceApportionmentId)`

uses all `SourceApportionment` rows having the same

`(CountryCode, SourceApportionmentId)`

SourceApportionment Primary Key

`CountryCode`
`SourceApportionmentId`
`ContributionType`
`SpatialScale`
`SourceSector`

Why No Physical FK Can Be Defined

`SourceApportionmentId` identifies an entire source-apportionment dataset containing several contribution rows.

`CompliancePlanLink` does not identify one individual contribution row and therefore does not contain:

`ContributionType`
`SpatialScale`
`SourceSector`
This is a **group relationship**.

PlanScenario ↔ ScenarioMeasure

Relationship

`ScenarioMeasure`

```
(CountryCode, ScenarioId, ScenarioCategory)
```

corresponds to

`PlanScenario`

```
(CountryCode, PlanId, ScenarioId, ScenarioCategory)
```

Why No Physical FK Can Be Defined

`ScenarioMeasure` does not contain `PlanId`, which forms part of the `PlanScenario` primary key.

The scenario-measure record identifies the scenario context but without pointing directly to a plan.

ScenarioMeasure → Measure

Relationship

`ScenarioMeasure`

```
(CountryCode, MeasureGroupId)
```

uses all `Measure` records having the same

```
(CountryCode, MeasureGroupId)
```

Measure Primary Key

`CountryCode`

`MeasureGroupId`

`MeasureId`

Why No Physical FK Can Be Defined

`ScenarioMeasure` points to a measure group included in a given scenario-category, not to one individual measure.

It does not contain `MeasureId`.

Conversely, `Measure` does not contain the scenario attributes required to refer to the complete `ScenarioMeasure` primary key.

This is a **group relationship**.

ScenarioMeasure → ModelObjectiveEstimation

Relationship

`ScenarioMeasure.MeasureReductionAssessmentMethodId`

identifies

`ModelObjectiveEstimation.AssessmentMethodId`

The method is used to estimate the reduction associated with the measure group.

Why No Physical FK Can Be Defined

`ScenarioMeasure` does not contain `DataAggregationProcessId`, which forms part of the complete primary key of `ModelObjectiveEstimation`.

Consequently, `ScenarioMeasure.MeasureReductionAssessmentMethodId` alone does not uniquely identify a corresponding `ModelObjectiveEstimation` record, and a conventional SQL foreign key cannot be defined.

Derivation and Use of DataAggregationProcessId

The missing `DataAggregationProcessId` can be derived indirectly through the associated `ScenarioId` in the `PlanScenario` table.

The relationship can be interpreted as follows:

```
ScenarioMeasure.ScenarioId
    → PlanScenario.ScenarioId

PlanScenario.DataAggregationProcessId
    → ModelObjectiveEstimation.DataAggregationProcessId
```

The derived `DataAggregationProcessId`, together with `MeasureReductionAssessmentMethodId`, provides the complete identifier required to locate the corresponding `ModelObjectiveEstimation` record.

Unlike the relationships involving `SpatialRepresentativeness` and `PollutionLevelAdjustment`, the aggregation process is expected to be fully consistent throughout this relationship.

Therefore, the `DataAggregationProcessId` derived through `PlanScenario` **should be identical** to the `DataAggregationProcessId` of the corresponding `AssessmentMethodId` in the `ModelObjectiveEstimation` table.

Any discrepancy should be considered an inconsistency in the reported data and should be reported as a validation error.

Although the relationship cannot be implemented as a conventional physical foreign key because `ScenarioMeasure` does not directly contain the complete parent primary key, the derivation described above provides a deterministic mapping that can be used for validation purposes.

Logical Relationship Summary

The following relationships cannot be represented correctly as conventional physical PK–FK constraints under the current reporting model.

Logical Relationship List

Authority

→ alternative entity tables
SamplingPointLocation

↔ ObservationMeasurementResult through validity periods
↔ ObservationMeasurementResultPNSD through validity periods
SamplingProcess

↔ ObservationMeasurementResult through validity periods
↔ ObservationMeasurementResultPNSD through validity periods
AssessmentRegimeZone

→ ZoneGeometry conditionally
ComplianceAssessmentMethod

→ SamplingPoint or ModelObjectiveEstimation
→ SpatialRepresentativeness group
← PollutionLevelAdjustment through AttainmentId
← CompliancePlanLink through AttainmentId
SpatialRepresentativeness

↔ SRSInline shared application dataset
↔ SRSExternal shared application dataset
→ ModelObjectiveEstimation through a method identifier
PollutionLevelAdjustment

→ ModelObjectiveEstimation through a partial model key
CompliancePlanLink

→ PlanScenario through a partial key
→ SourceApportionment group
PlanScenario

↔ ScenarioMeasure through a partial key

ScenarioMeasure

→ Measure group
→ ModelObjectiveEstimation through a partial model key
Documentation

↔ reporting-table documentation attributes using `DataTable + DocumentType + DocumentId`

Documentation Relationships

Documentation relationships are contextual relationships between the `Documentation` table and documentation-reference attributes contained in different reporting tables.

The applicable reporting table and documentation-reference attribute are determined by the combination of:

`DataTable`;

`DocumentType`;

`DocumentId`;

`CountryCode`.

Documentation Primary Key

The `Documentation` table is identified by a contextual composite key.

Documentation Primary Key

```
ALTER TABLE [reporting].[Documentation]
ADD CONSTRAINT [PK_Documentation]
PRIMARY KEY
(
    [CountryCode],
    [DataTable],
    [DocumentType],
    [DocumentId]
);
```

Meaning of DataTable

`Documentation.DataTable` indicates the reporting table in which a process must look for the documentation identifier.

For example:

```
DataTable = 'ModelObjectiveEstimation'
```

means that the `DocumentId` must be found in a documentation-reference column in:

```
[reporting].[ModelObjectiveEstimation]
```

Similarly:

```
DataTable = 'SamplingProcess'
```

means that the document identifier must be looked up in:

```
[reporting].[SamplingProcess]
```

`DataTable` therefore identifies the reporting-table context of the documentation reference.

Meaning of DocumentType

`Documentation.DocumentType` identifies which documentation-reference attribute in the specified reporting table contains the `DocumentId`.

For example, `ModelObjectiveEstimation` may contain:

`MethodDocumentId`;
`DataQualityDocumentId`.
A documentation row with:

```
DataTable = 'ModelObjectiveEstimation'
DocumentType = 'MethodDocument'
DocumentId = 'DOC-001'
```

means that a process should search for `DOC-001` in:

```
ModelObjectiveEstimation.MethodDocumentId
```

A documentation row with:

```
DataTable = 'ModelObjectiveEstimation'
DocumentType = 'DataQualityDocument'
DocumentId = 'DOC-002'
```

means that a process should search for `DOC-002` in:

```
ModelObjectiveEstimation.DataQualityDocumentId
```

`DocumentType` therefore distinguishes between different documentation-reference attributes within the same reporting table.

Documentation Lookup Logic

The relationship is not simply:

```
Child.DocumentId → Documentation.DocumentId
```

Instead, the lookup works as follows:

```
Documentation.DataTable
    identifies the reporting table

Documentation.DocumentType
    identifies the documentation-reference attribute in that table

Documentation.DocumentId
    is matched against the value reported in that attribute
```

Conceptually:

```
Documentation row
    (CountryCode, DataTable, DocumentType, DocumentId)

maps to

the row or rows in the indicated reporting table
where the documentation column selected by DocumentType
contains the same DocumentId
```

Example — ModelObjectiveEstimation Documentation

Assume that `ModelObjectiveEstimation` contains:

`MethodDocumentId`;

`DataQualityDocumentId`.

Method Document

```
Documentation.DataTable
    = 'ModelObjectiveEstimation'

Documentation.DocumentType
    = 'MethodDocument'

Documentation.DocumentId
    is searched in
    ModelObjectiveEstimation.MethodDocumentId
```

Data-Quality Document

```
Documentation.DataTable
    = 'ModelObjectiveEstimation'

Documentation.DocumentType
    = 'DataQualityDocument'

Documentation.DocumentId
    is searched in
    ModelObjectiveEstimation.DataQualityDocumentId
```

The combination of `DataTable` and `DocumentType` determines which reporting table and which documentation-reference column must contain the identifier.

Example — SamplingProcess Documentation

Assume that `SamplingProcess` contains:

`DataQualityDocumentId`;
`EquivalenceDemonstrationDocumentId`;
`ProcessDocumentId`.

The corresponding mappings are conceptually:

```
DataTable = 'SamplingProcess'
DocumentType = 'DataQualityDocument'
    → SamplingProcess.DataQualityDocumentId

DataTable = 'SamplingProcess'
DocumentType = 'EquivalenceDemonstrationDocument'
    → SamplingProcess.EquivalenceDemonstrationDocumentId

DataTable = 'SamplingProcess'
DocumentType = 'ProcessDocument'
    → SamplingProcess.ProcessDocumentId
```

Each `DocumentType` value selects a different documentation-reference attribute in the same reporting table.

Example — MeasurementStation Documentation

If `MeasurementStation` contains:

```
NetworkDocumentId
```

the corresponding logical mapping is:

```
DataTable = 'MeasurementStation'
DocumentType = 'NetworkDocument'
    → MeasurementStation.NetworkDocumentId
```

Example — PollutionLevelAdjustment Documentation

If `PollutionLevelAdjustment` contains:

```
AdjustmentDocumentId
```

the corresponding mapping is:

```
DataTable = 'PollutionLevelAdjustment'
DocumentType = 'AdjustmentDocument'
    → PollutionLevelAdjustment.AdjustmentDocumentId
```

Example — CompliancePlanLink Documentation

If `CompliancePlanLink` contains:

```
PlanDocumentId
```

the corresponding mapping is:

```
DataTable = 'CompliancePlanLink'
DocumentType = 'PlanDocument'
    → CompliancePlanLink.PlanDocumentId
```

Example — SourceApportionment Documentation

If `SourceApportionment` contains:

```
SourceApportionmentDocumentId
```

the corresponding mapping is:

```
DataTable = 'SourceApportionment'
DocumentType = 'SourceApportionmentDocument'
    → SourceApportionment.SourceApportionmentDocumentId
```

Direction of the Documentation Relationship

The documentation relationship can be interpreted from either direction.

From a Reporting Table to Documentation

A process reads a documentation-reference attribute, for example:

```
ModelObjectiveEstimation.MethodDocumentId = 'DOC-001'
```

It then expects a `Documentation` row containing:

```
CountryCode = the same country
DataTable = 'ModelObjectiveEstimation'
DocumentType = 'MethodDocument'
DocumentId = 'DOC-001'
```

From Documentation to a Reporting Table

A process reads the following values from the `Documentation` row:

```
CountryCode
DataTable
DocumentType
DocumentId
```

It then uses:

`DataTable` to select the reporting table;
`DocumentType` to select the documentation-reference attribute;
`DocumentId` to find the matching value;
`CountryCode` to restrict the lookup to the appropriate country.

Why Documentation Relationships Are Not Physical FKs

Reporting tables normally contain:

```
CountryCode
a specific document identifier attribute
```

For example:

```
CountryCode
MethodDocumentId
```

or:

```
CountryCode
DataQualityDocumentId
```

The reporting rows do not physically contain:

```
DataTable
DocumentType
```

Those values are implied by:

the reporting table in which the document identifier appears;
the particular document-reference attribute being used.
A normal FK from:

```
ModelObjectiveEstimation
    (CountryCode, MethodDocumentId)
```

to:

```
Documentation
    (CountryCode, DataTable, DocumentType, DocumentId)
```

cannot be defined because the child does not contain the complete `Documentation` PK.

It would also be incorrect to reference only:

```
CountryCode + DocumentId
```

because the same `DocumentId` could potentially occur in different:

table contexts;
document-type contexts.
The relationship is therefore contextual and must be implemented through processing or QC validation.

Documentation Mapping Specification

A mapping configuration may be used by the processing logic to define which `DocumentType` corresponds to which identifier attribute:

MeasurementStation

Network → `NetworkDocumentId`
SamplingProcess

DataQuality → `DataQualityDocumentId`
EquivalenceDemonstration → `EquivalenceDemonstrationDocumentId`
Process → `ProcessDocumentId`
ModelObjectiveEstimation

Method → `MethodDocumentId`
DataQuality → `DataQualityDocumentId`
PollutionLevelAdjustment

Adjustment → `AdjustmentDocumentId`
CompliancePlanLink

Plan → `PlanDocumentId`
SourceApportionment

SourceApportionment → `SourceApportionmentDocumentId`
The exact vocabulary values used in `DataTable` and `DocumentType` must correspond to the values defined in the reporting model.

Any additional documentation-reference attributes should be added to the mapping using the same principle.

Summary

The SQL specification distinguishes between physical referential constraints and logical relationships.

Physical foreign keys are proposed only where a child table contains the complete documented primary key of one uniquely identifiable parent row and where the relationship applies unconditionally.

Other documented relationships cannot be represented correctly using conventional SQL foreign keys under the current reporting model.

These include:

temporal relationships;

alternative-parent relationships;

group-level references;

shared result datasets;

partial-key method references;

conditional geographic relationships;

contextual documentation references.

For documentation records:

`DataTable` identifies the reporting table in which the documentation identifier must be located;

`DocumentType` identifies the particular documentation-reference attribute within that table;

`DocumentId` contains the identifier being matched;

`CountryCode` restricts the relationship to the applicable country.

The absence of a physical FK for a relationship does not mean that the tables are unrelated.

It means that the relationship contains additional business semantics that cannot be expressed through an ordinary equality-based PK–FK constraint without changing the existing reporting-table design.